

The Tempo Mode Matrix Press **Tempo** to access the following tempo parameters:

Simultaneously press **Up** and **Down** to return to 0.0.

0 Tempo	0.0 Rate	0.1 Beat Value	0.2 Source	
1 Tap	1.0 Duration	1.1 Source	1.2 Average	1.3 Display

An asterisk (*) accompanying a parameter name indicates that there are subparameters available at that matrix location. The **Load /*** LED will light whenever an asterisk appears in the display. Press **Load /*** to step to the next subparameter.

From any point in the matrix, press **Up** or **Down** together with **Load /*** to backstep to the previous parameter.

Row 0 Tempo

0 Tempo	0.0 Rate	0.1 Beat Value	0.2 Source
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0.0 Rate

This is the current tempo (in Beats Per Minute). When 0.2 is set to **Internal**, turning ADJUST allows you to select a different rate (40-400 BPM). The Tempo LED will flash at the new rate. Note that fractional tempos can be tapped in, but are not available via ADJUST. The display will always show the nearest whole number value.

0.1 BeatValue

Tempo is expressed in BPM. This control allows you to specify the value of the beat. Eighth, dotted-eighth, quarter, dotted-quarter, half, dotted-half, and whole-note values are available. If, for example, the rate is 120 BPM, and you select eighth-note here, the tempo will be 120 eighth-notes per minute. If you select quarter-note here, the tempo will be 120 quarter-notes per minute. (The factory default is quarter-note.)

0.2 Source

You can choose to have tempo determined by the PCM 81 **Tap** and **Rate** controls (Internal), or by MIDI Clock. When MIDI Clock is selected as the tempo source, **Tap** acts as a reset, setting the downbeat of the LFO and the time-based switches.

1 Tap	1.0 Duration	1.1 Source	1.2 Average	1.3 Display
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**Row 1
Tap**

1.0 Tap Duration

This control determines how many beats will occur in a tap interval. The default setting (1 beat) is probably adequate for most applications. With the default setting of 1 beat, if the tempo is set to 120 bpm, and the beat value is set to quarter-note, each TAP = 1 quarter-note = 1 beat. Available values are: 1/8, 1/7, 1/6, 1/5, 1/4, 1/3, 1/2, 1-8 beats

1.1 Tap Source and Tap Level

Press **Load/★** to toggle between these two controls.

Tap Source allows you to assign the Tap function to any of the PCM 81's Internal, MIDI, or MIDI controllers as listed under Patching.

Tap Level allows you to set the level at which the Tap function is triggered.

1.2 Average

This control allows you to average the last 2-8 taps. Higher numbers mean that the response to incoming taps will be more gradual. (The tempo is updated on every tap, but with a value which is the average of the last 2-8 taps.) Higher average values are more useful if you're trying to lock into a pre-recorded track.

1.3 Display

This control allows you to disable the flashing of the Tempo LED. Turn ADJUST to select **Off**. The Tempo LED will turn off.

Editing an Effect

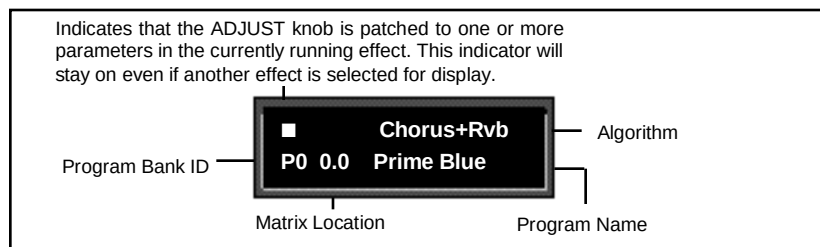
With 17 algorithms and 300 preset effects, the PCM 81 gives you a lot to play with right out of the box. An enormous range of editing control is provided for each algorithm, with parameters organized in an edit matrix of as many as 100 main controls. In addition to providing this powerful sound design capability, the PCM 81 is also designed to allow you to customize these controls for your day-to-day editing needs, or to simply use a subset of controls specially designed for each preset.

This section will describe three basic levels of editing, from the simplest "plug and play" method, through the full edit matrix.

The Soft Knob

Every preset in the PCM 81 has one or more of its available parameters patched to the front panel ADJUST knob. This Soft Knob provides the first level of editing control. Once you have loaded a preset, simply turn ADJUST to alter the effect.

When shipped, the PCM 81 will power up with the first preset (**P0 0.0 Prime Blue**) loaded. The following display will appear:

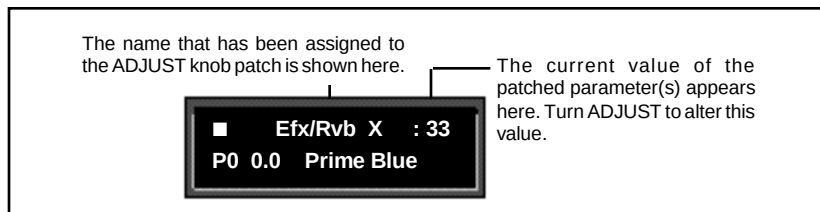


Turn ADJUST. The display will change to show the name assigned to the ADJUST patch, and the current value of the patch. Continue turning ADJUST to change the value of the patch along its entire range. The range of ADJUST knob control can be limited in Control Mode, making it possible, for example, to have a range of only 0...1, in order to have the ADJUST knob behave as an Off/On control. Many of the presets have range limits to make them more convenient to use. In some of the presets, the changes effected by ADJUST will be as simple as altering the wet/dry mix. In others, such as the example shown below, turning ADJUST will glide you all the way from a chorusing effect to a reverb effect over the complete control range of 0-127.

Controlling the Soft Knob with a Foot Pedal

If you have a foot pedal connected to the PCM rear panel Foot Controller jack, you can use it to control the soft knob patch. (Note that no MIDI connections are required to do this.)

To assign a foot pedal control over the soft knob patches, set both Control mode 3.5 **ADJUST** and Control mode 3.5 **Foot Pedal** to the same MIDI Controller. See Chapter 5 *MIDI Operation*.



Details on how to create your own ADJUST knob patches are given at the end of this chapter under *Patching*.

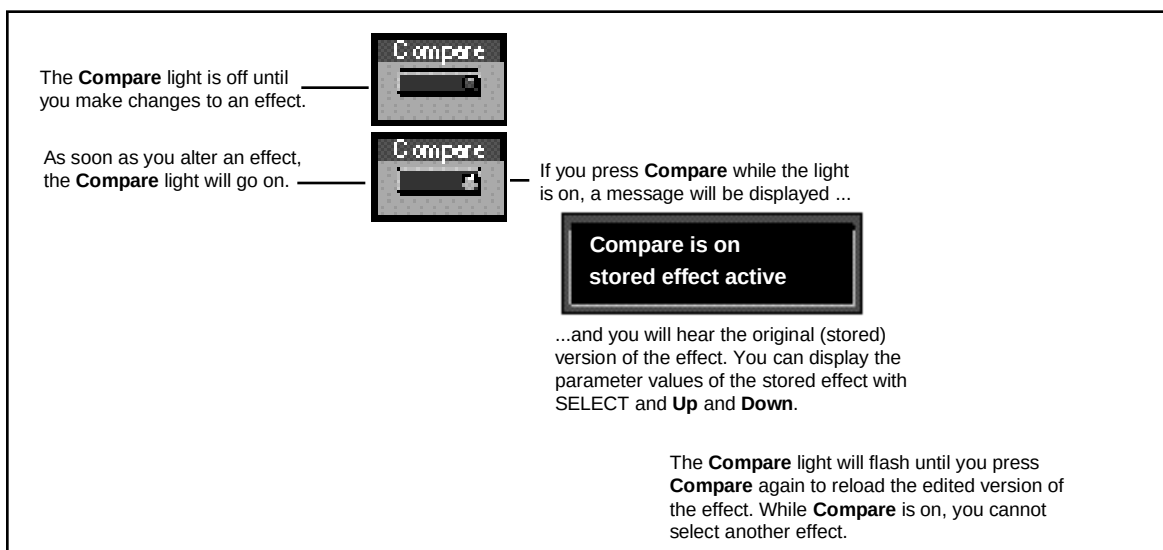
Turn SELECT to scroll all of the available parameters in the Soft Row. Turn ADJUST to change the value of any displayed parameter along its entire range.



Compare Whenever you edit a PCM 81 effect from the front panel, the LED in the **Compare** button will light. This lets you know that the effect has been altered since the last store operation, and that the edit compare function is active. Whenever this light is on, you can press **Compare** to hear the original version of the effect. A message will be displayed to inform you that the stored version of the effect is being loaded. Although this message is only displayed briefly, the **Compare** LED will flash to let you know the effect you are hearing is the stored version. While **Compare** is on, you can use the SELECT knob and the **Up** and **Down** buttons to view parameter values in the stored effect.

Press **Compare** again to reload your edited version. Another message will be displayed, and the **Compare** LED will stop flashing and remain lit until you store your edited version, or select and load another effect.

Note: Altering parameter values with patched sources other than ADJUST will not light the LED.



Bypass The front panel **Bypass** button is always active, and will turn on the type of bypass (**AllMute**, **InputMute**, **OutputMute**, or **Bypass**) selected in Control Mode 1.3.

When you press **Bypass**, the button LED will light and a message will be displayed to inform you that bypass is on. The display message will also indicate the type of bypass which is in effect. Press **Bypass** again to turn both the LED and the bypass condition off. A brief display message will inform you that bypass is off.

Press and hold **Bypass** to display the current state and type of bypass.

Store Operations

The PCM 81 is shipped with its Memory Protection option on. When you press the front panel **Store** button, the following message will be displayed:

Store not active
Mem Protect is on

To turn Memory Protection off, press **Control** and use **Up** and **Down** and **SELECT** to display matrix location 1.5. The display should read:

System Mem Protect
1.5 On

Turn **ADJUST** to select **Off**.

Turning Memory Protection Off

With Memory Protection disabled, pressing **Store** will cause a message like the following one to be displayed:

An asterisk (★) is positioned in front of the display label which is available for changing — in this case, the name of the effect.

The default Bank and register location will appear here.

Name: ★Prime Blue
R 0.0 available

"available" means that this location is empty. If an effect is stored at this location, its name will appear here.

Press **Load/★** to select the **ADJUST** patch for renaming. Press again to select the register bank. Press again to select the register location within the selected bank.

Storing an Effect

If you press **Store** again, the display will ask "Are you sure?" (Press **STORE**). Pressing **Store** will cause the currently running effect to be saved, as is, in the location shown in the lower lefthand corner of the display. A brief message will inform you that the effect is being stored, then the PCM 81 will revert to the mode it was in before **Store** was pressed, with the newly-stored effect loaded.

When you store an effect, the following things are saved:

- Values of all Edit matrix parameters. This includes the initial values for any patch destinations when **Patch Update** (Control mode 1.7) is set to **Delayed**.
- Soft Row parameter assignments.
- Value of **ADJUST** when the effect was stored.
- Last parameter selected when the effect was stored.
- Tempo **Rate** and **BeatValue**.
- Tap **Duration** and **Average**.
- **ADJUST** name.
- Effect name.

How the PCM 81 selects a default Bank and Register

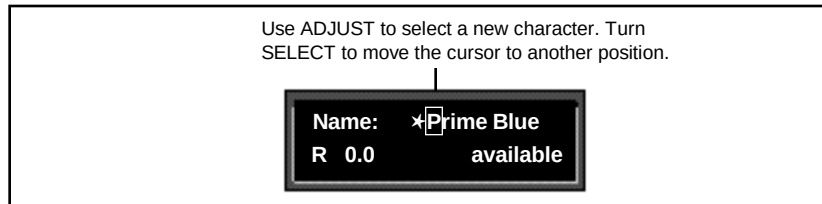
If a register is loaded when **Store** is pressed, the default location will be the same as the loaded effect. If a preset is loaded when **Store** is pressed, the default location is determined as follows:

The PCM 81 always uses the last-stored register location as a starting point. Its default choice is the next "available" register within that bank. If there are no available registers at a higher location number within the same bank, the default location will be that of the last stored register.

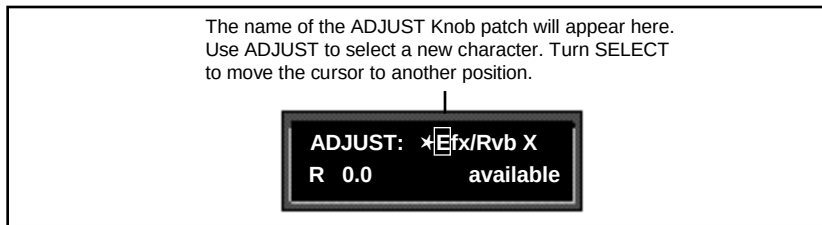
If the last stored register is on a RAM card, and the RAM card is not inserted, the PCM 81 will search for available registers starting at 0.0 in the Internal Register Bank.

Renaming the Effect

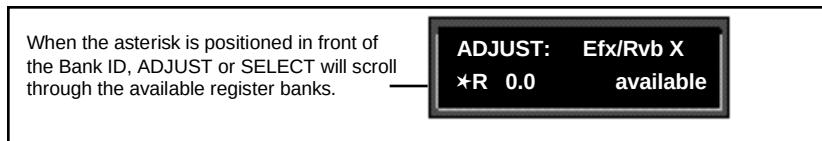
Renaming an effect is straightforward. With the asterisk and the cursor positioned as shown, turn ADJUST to select a new character. Press **Up** or **Down** to select a new *type* of character (upper case, lower case, numeric, symbolic, or blank). Simultaneously press **Up** and **Down** to clear all characters from the cursor to the end of the line. Turn SELECT to reposition the cursor over another character, and use ADJUST to change it. Continue in this manner until you have finished entering the new name. A maximum of 12 characters (including spaces) may be used.

**Renaming the ADJUST Knob Patch**

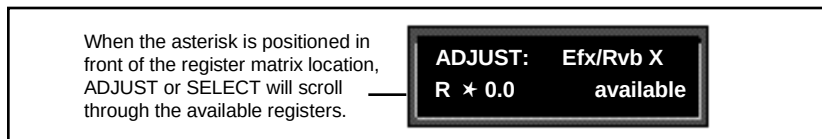
Press **Load/*** to select the ADJUST patch for renaming. The display will change to show the following:

**Selecting a Bank and Register Location**

Press **Load/*** to move the asterisk to the Register Bank ID.



Press **Load/*** again to move the asterisk to the register matrix location.



Once you have made all the changes you want to an effect, and have selected a register location, press **Store**. If you press **Store**, the display will ask "Are you sure?" (Press STORE). If you don't want to save the effect as shown, press any button to cancel the operation. Press **Store** to complete the operation.

Setting Edit Mode to **Pro** (Control Mode 1.0) gives you access to the full parameter matrix of the algorithm for any loaded effect whenever you press the front panel **Edit** button.

The Full Edit Matrix

To select any parameter for adjustment, use **SELECT** to move horizontally across the matrix and the **Up** and **Down** buttons to move vertically. An asterisk (*) appearing before a displayed parameter indicates that more than one parameter is available at that location. Press **Load/★** to display these additional parameters.

Each of the 17 PCM 81 algorithms has a unique matrix, but many parameters are common to all effects, and their placement within the matrix is consistent. For example, all parameters within a given row of any matrix are related. This type of grouping is immediately apparent from the name of the row. For example the "Levels" row in any matrix will contain all of the available level controls for the effect's delay voices.

As far as possible, rows with similar controls will always appear in the same position in the matrix and, within each row, parameters will generally appear in the same position. For example, the first row (0) in every algorithm is "Controls". The **Mix** parameter, which is available in all effects, is always located in position 0 in this row. Similarly, the last two rows of every matrix are "Modulation" and "Patches". This makes it very easy to find your way around the large number of available parameters, and to anticipate where to find the controls you are looking for when you switch between effects.

Concert Hall									
0 Controls	0.0 Mix	0.1 FX Adjw	0.2 *Rel Lvl Pan	0.3 High Cut	0.4 Voice Dtl	0.5 FX Mix	0.6 FX Width	0.7 *RCU U6T	
1 Prob Time	1.0 Low R	1.1 Mid R	1.2 Crossover	1.3 R H C	1.4 Pre Dela.	1.5 *Rel Lvl DL	1.6 *Ph GM		
2 Prob Design	2.0 Size								
Plate									
3 Levels	3.0 Master	0 Controls	0.0 Mix	0.1 FX Adjw	0.2 *Rel Lvl Pan	0.3 High Cut	0.4 Voice Dtl	0.5 FX Mix	0.6 FX Width
4 Delay Time	4.0 *Master	1 Prob Time	1.0 Low R	1.1 Mid R	1.2 Crossover	1.3 R H C	1.4 Pre Dela.	1.5 *Rel Lvl DL	1.6 *Ph GM
5 Feedback	5.0 Master	2 Prob Design	2.0 Size						
6 Panning	6.0 Master	3 Levels	3.0 Master						
7 Modulation	7.0 Mod LFO	4 Delay Time	4.0 *Master						
Patch++	Patch 0	5 Feedback	5.0 Master						
		6 Panning	6.0 Master						
		7 Modulation	7.0 Mod LFO						
		Patch++	Patch 0						
M-Band+Rvb									
0 Controls	0.0 Mix	0.1 FX Adjw	0.2 *Rel Lvl Pan	0.3 FX Mix	0.4 FX Width	0.5 *RCU U6T			
1 Prob Time	1.0 Low R	1.1 Mid R	1.2 Crossover	1.3 R H C	1.4 Pre Dela.	1.5 *Rel Lvl DL	1.6 *Ph GM		
2 Prob Design	2.0 Size	2.1 Diffusion	2.2 Shape	2.3 Spread	2.4 Spin	2.5 Lnk	2.6 Prob Out		
3 Levels	3.0 Master	3.1 Voice 1	3.2 Voice 2	3.3 Voice 3	3.4 Voice 4	3.5 Voice 5	3.6 Voice 6		
4 Delay Time	4.0 *Master	4.1 Voice 1	4.2 Voice 2	4.3 Voice 3	4.4 Voice 4	4.5 Voice 5	4.6 Voice 6		
5 Feedback	5.0 Master	5.1 Voice 1	5.2 Voice 2	5.3 Voice 3	5.4 Voice 4	5.5 Voice 5	5.6 Voice 6		
6 Panning	6.0 Master	6.1 Voice 1	6.2 Voice 2	6.3 Voice 3	6.4 Voice 4	6.5 Voice 5	6.6 Voice 6		
7 Modulation	7.0 Master	7.1 Voice 1	7.2 Voice 2	7.3 Voice 3	7.4 Voice 4	7.5 Voice 5	7.6 Voice 6		
8 Mod LFO	8.0 Mod LFO	8.1 Mod RFR Env	8.2 Mod SFR L	8.3 Mod SFR R	8.4 Mod Lach	8.5 Mod SFR L	8.6 Mod SFR R		
Patch++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8

Similar parameters can be found in similar positions in every effect matrix. Once you are familiar with one edit matrix, you will find it easy to anticipate where to find controls in another matrix.

Detailed information about each individual edit matrix, as well as parameter and effect descriptions are given in Chapter 3.

Creating a Soft Row In **Pro** mode you still have complete access to the Soft Row, which appears above row 0 of the full edit matrix. The parameters assigned here are duplicates of selected parameters in the matrix and can be adjusted from Row **S** (Soft Row), or from their matrix location.

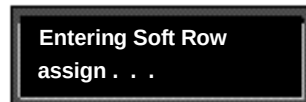
The following example shows the edit matrix for the preset, **P 0.0 Prime Blue** (Chorus+Rvb algorithm). As you can see, parameters from locations throughout the matrix have been assigned to this Soft Row. In this preset (and in every preset) we have designed the Soft Row by assigning the parameters in numerical order, so that parameters from the Controls row are assigned in order to the first locations, followed by assignments from the Rvb Time row, and so on. As Soft Row parameters are also labeled with the name of the row from which they were taken, this makes it relatively easy to find the source parameter for any Soft Row assignment.

The Soft Row assigned to an effect appears
above Row 0 in the full Edit matrix.

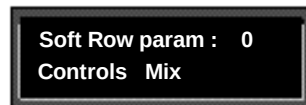
# (Full Row)	Control C-1 to C-8	Control C-9 to C-16	Control C-17 to C-24	Env. Time E-1 to E-8	Env. Time E-9 to E-16	Env. Time E-17 to E-24	Env. Time E-25 to E-32	Env. Time E-33 to E-40	Env. Time E-41 to E-48	Env. Time E-49 to E-56	Env. Time E-57 to E-64
0 Control	C-1	C-9	C-17	E-1	E-9	E-17	E-25	E-33	E-41	E-49	E-57
1 Env. Time	E-2	E-10	E-18	E-3	E-11	E-19	E-27	E-35	E-43	E-51	E-59
2 Env. Time	E-4	E-12	E-20	E-5	E-13	E-21	E-29	E-37	E-45	E-53	E-61
3 Env. Time	E-6	E-14	E-22	E-7	E-15	E-23	E-31	E-39	E-47	E-55	E-63
4 Env. Time	E-8	E-16	E-24	E-9	E-17	E-25	E-33	E-41	E-49	E-57	E-65
5 Env. Time	E-10	E-18	E-26	E-11	E-19	E-27	E-35	E-43	E-51	E-59	E-67
6 Env. Time	E-12	E-20	E-28	E-13	E-21	E-29	E-37	E-45	E-53	E-61	E-69
7 Env. Time	E-14	E-22	E-30	E-15	E-23	E-31	E-39	E-47	E-55	E-63	E-71
8 Env. Time	E-16	E-24	E-32	E-17	E-25	E-33	E-41	E-49	E-57	E-65	E-73
9 Env. Time	E-18	E-26	E-34	E-19	E-27	E-35	E-43	E-51	E-59	E-67	E-75
10 Env. Time	E-20	E-28	E-36	E-21	E-29	E-37	E-45	E-53	E-61	E-69	E-77
11 Env. Time	E-22	E-30	E-38	E-23	E-31	E-39	E-47	E-55	E-63	E-71	E-79
12 Env. Time	E-24	E-32	E-40	E-25	E-33	E-41	E-49	E-57	E-65	E-73	E-81
13 Env. Time	E-26	E-34	E-42	E-27	E-35	E-43	E-51	E-59	E-67	E-75	E-83
14 Env. Time	E-28	E-36	E-44	E-29	E-37	E-45	E-53	E-61	E-69	E-77	E-85
15 Env. Time	E-30	E-38	E-46	E-31	E-39	E-47	E-55	E-63	E-71	E-79	E-87
16 Env. Time	E-32	E-40	E-48	E-33	E-41	E-49	E-57	E-65	E-73	E-81	E-89
17 Env. Time	E-34	E-42	E-50	E-35	E-43	E-51	E-59	E-67	E-75	E-83	E-91
18 Env. Time	E-36	E-44	E-52	E-37	E-45	E-53	E-61	E-69	E-77	E-85	E-93
19 Env. Time	E-38	E-46	E-54	E-39	E-47	E-55	E-63	E-71	E-79	E-87	E-95
20 Env. Time	E-40	E-48	E-56	E-41	E-49	E-57	E-65	E-73	E-81	E-89	E-97
21 Env. Time	E-42	E-50	E-58	E-43	E-51	E-59	E-67	E-75	E-83	E-91	E-99
22 Env. Time	E-44	E-52	E-60	E-45	E-53	E-61	E-69	E-77	E-85	E-93	E-101
23 Env. Time	E-46	E-54	E-62	E-47	E-55	E-63	E-71	E-79	E-87	E-95	E-103
24 Env. Time	E-48	E-56	E-64	E-49	E-57	E-65	E-73	E-81	E-89	E-97	E-105
25 Env. Time	E-50	E-58	E-66	E-51	E-59	E-67	E-75	E-83	E-91	E-99	E-107
26 Env. Time	E-52	E-60	E-68	E-53	E-61	E-69	E-77	E-85	E-93	E-101	E-109
27 Env. Time	E-54	E-62	E-70	E-55	E-63	E-71	E-79	E-87	E-95	E-103	E-111
28 Env. Time	E-56	E-64	E-72	E-57	E-65	E-73	E-81	E-89	E-97	E-105	E-113
29 Env. Time	E-58	E-66	E-74	E-59	E-67	E-75	E-83	E-91	E-99	E-107	E-115
30 Env. Time	E-60	E-68	E-76	E-61	E-69	E-77	E-85	E-93	E-101	E-109	E-117
31 Env. Time	E-62	E-70	E-78	E-63	E-71	E-79	E-87	E-95	E-103	E-111	E-119
32 Env. Time	E-64	E-72	E-80	E-65	E-73	E-81	E-89	E-97	E-105	E-113	E-121
33 Env. Time	E-66	E-74	E-82	E-67	E-75	E-83	E-91	E-99	E-107	E-115	E-123
34 Env. Time	E-68	E-76	E-84	E-69	E-77	E-85	E-93	E-101	E-109	E-117	E-125
35 Env. Time	E-70	E-78	E-86	E-71	E-79	E-87	E-95	E-103	E-111	E-119	E-127
36 Env. Time	E-72	E-80	E-88	E-73	E-81	E-89	E-97	E-105	E-113	E-121	E-129
37 Env. Time	E-74	E-82	E-90	E-75	E-83	E-91	E-99	E-107	E-115	E-123	E-131
38 Env. Time	E-76	E-84	E-92	E-77	E-85	E-93	E-101	E-109	E-117	E-125	E-133
39 Env. Time	E-78	E-86	E-94	E-79	E-87	E-95	E-103	E-111	E-119	E-127	E-135
40 Env. Time	E-80	E-88	E-96	E-81	E-89	E-97	E-105	E-113	E-121	E-129	E-137
41 Env. Time	E-82	E-90	E-98	E-83	E-91	E-99	E-107	E-115	E-123	E-131	E-139
42 Env. Time	E-84	E-92	E-100	E-85	E-93	E-101	E-109	E-117	E-125	E-133	E-141
43 Env. Time	E-86	E-94	E-102	E-87	E-95	E-103	E-111	E-119	E-127	E-135	E-143
44 Env. Time	E-88	E-96	E-104	E-89	E-97	E-105	E-113	E-121	E-129	E-137	E-145
45 Env. Time	E-90	E-98	E-106	E-91	E-99	E-107	E-115	E-123	E-131	E-139	E-147
46 Env. Time	E-92	E-100	E-108	E-93	E-101	E-109	E-117	E-125	E-133	E-141	E-149
47 Env. Time	E-94	E-102	E-110	E-95	E-103	E-111	E-119	E-127	E-135	E-143	E-151
48 Env. Time	E-96	E-104	E-112	E-97	E-105	E-113	E-121	E-129	E-137	E-145	E-153
49 Env. Time	E-98	E-106	E-114	E-99	E-107	E-115	E-123	E-131	E-139	E-147	E-155
50 Env. Time	E-100	E-108	E-116	E-101	E-109	E-117	E-125	E-133	E-141	E-149	E-157
51 Env. Time	E-102	E-110	E-118	E-103	E-111	E-119	E-127	E-135	E-143	E-151	E-159
52 Env. Time	E-104	E-112	E-120	E-105	E-113	E-121	E-129	E-137	E-145	E-153	E-161
53 Env. Time	E-106	E-114	E-122	E-107	E-115	E-123	E-131	E-139	E-147	E-155	E-163
54 Env. Time	E-108	E-116	E-124	E-109	E-117	E-125	E-133	E-141	E-149	E-157	E-165
55 Env. Time	E-110	E-118	E-126	E-111	E-119	E-127	E-135	E-143	E-151	E-159	E-167
56 Env. Time	E-112	E-120	E-128	E-113	E-121	E-129	E-137	E-145	E-153	E-161	E-169
57 Env. Time	E-114	E-122	E-130	E-115	E-123	E-131	E-139	E-147	E-155	E-163	E-171
58 Env. Time	E-116	E-124	E-132	E-117	E-125	E-133	E-141	E-149	E-157	E-165	E-173
59 Env. Time	E-118	E-126	E-134	E-119	E-127	E-135	E-143	E-151	E-159	E-167	E-175
60 Env. Time	E-120	E-128	E-136	E-121	E-129	E-137	E-145	E-153	E-161	E-169	E-177
61 Env. Time	E-122	E-130	E-138	E-123	E-131	E-139	E-147	E-155	E-163	E-171	E-179
62 Env. Time	E-124	E-132	E-140	E-125	E-133	E-141	E-149	E-157	E-165	E-173	E-181
63 Env. Time	E-126	E-134	E-142	E-127	E-135	E-143	E-151	E-159	E-167	E-175	E-183
64 Env. Time	E-128	E-136	E-144	E-129	E-137	E-145	E-153	E-161	E-169	E-177	E-185
65 Env. Time	E-130	E-138	E-146	E-131	E-139	E-147	E-155	E-163	E-171	E-179	E-187
66 Env. Time	E-132	E-140	E-148	E-133	E-141	E-149	E-157	E-165	E-173	E-181	E-189
67 Env. Time	E-134	E-142	E-150	E-135	E-143	E-151	E-159	E-167	E-175	E-183	E-191
68 Env. Time	E-136	E-144	E-152	E-137	E-145	E-153	E-161	E-169	E-177	E-185	E-193
69 Env. Time	E-138	E-146	E-154	E-139	E-147	E-155	E-163	E-171	E-179	E-187	E-195
70 Env. Time	E-140	E-148	E-156	E-141	E-149	E-157	E-165	E-173	E-181	E-189	E-197
71 Env. Time	E-142	E-150	E-158	E-143	E-151	E-159	E-167	E-175	E-183	E-191	E-199
72 Env. Time	E-144	E-152	E-160	E-145	E-153	E-161	E-169	E-177	E-185	E-193	E-201
73 Env. Time	E-146	E-154	E-162	E-147	E-155	E-163	E-171	E-179	E-187	E-195	E-203
74 Env. Time	E-148	E-156	E-164	E-149	E-157	E-165	E-173	E-181	E-189	E-197	E-205
75 Env. Time	E-150	E-158	E-166	E-151	E-159	E-167	E-175	E-183	E-191	E-199	E-207
76 Env. Time	E-152	E-160	E-168	E-153	E-161	E-169	E-177	E-185	E-193	E-201	E-209
77 Env. Time	E-154	E-162	E-170	E-155	E-163	E-171	E-179	E-187	E-195	E-203	E-211
78 Env. Time	E-156	E-164	E-172	E-157	E-165	E-173	E-181	E-189	E-197	E-205	E-213
79 Env. Time	E-158	E-166	E-174	E-159	E-167	E-175	E-183	E-191	E-199	E-207	E-215
80 Env. Time	E-160	E-168	E-176	E-161	E-169	E-177	E-185	E-193	E-201	E-209	E-217
81 Env. Time	E-162	E-170	E-178	E-163	E-171	E-179	E-187	E-195	E-203	E-211	E-219
82 Env. Time	E-164	E-172	E-180	E-165	E-173	E-181	E-189	E-197	E-205	E-213	E-221
83 Env. Time	E-166	E-174	E-182	E-167	E-175	E-183	E-191	E-199	E-207	E-215	E-223
84 Env. Time	E-168	E-176	E-184	E-169	E-177	E-185	E-193	E-201	E-209	E-217	E-225
85 Env. Time	E-170	E-178	E-186	E-171	E-179	E-187	E-195	E-203	E-211	E-219	E-227
86 Env. Time	E-172	E-180	E-188	E-173	E-181	E-189	E-197	E-205	E-213	E-221	E-229
87 Env. Time	E-174	E-182	E-190	E-175	E-183	E-191	E-199	E-207	E-215	E-223	E-231
88 Env. Time	E-176	E-184	E-192	E-177	E-185	E-193	E-201	E-209	E-217	E-225	E-233
89 Env. Time	E-178	E-186	E-194	E-179	E-187	E-195	E-203	E-211	E-219	E-227	E-235
90 Env. Time	E-180	E-188	E-196	E-181	E-189	E-197	E-205	E-213	E-221	E-229	E-237
91 Env. Time	E-182	E-190	E-198	E-183	E-191	E-199	E-207	E-215	E-223	E-231	E-239
92 Env. Time	E-184	E-192	E-200	E-185	E-193	E-201	E-209	E-217	E-225	E-233	E-241
93 Env. Time	E-186	E-194	E-202	E-187	E-195	E-203	E-211	E			

Modifying the Soft Row, or creating a completely new Soft Row for an effect is easy:

1. From the full Edit matrix, press **Up** until you get to the Soft Row, indicated by an **S** in the lower left corner of the display.
2. With any Soft Row parameter displayed, press and hold down the **Edit** button. The following display will appear briefly.



When you release the **Edit** button, the display will change to the Soft Row Assignment display shown below. The **Edit** button LED will begin flashing and will continue to flash as long as you are in Soft Row Assign.



Here we show the first Soft Row parameter in our example — the actual Soft Row parameter selected will correspond to whichever one was displayed when you pressed and held **Edit**.

3. Turn ADJUST to scroll through all of the available parameters in the Edit matrix row by row, in numerical order. The entry "available" can also be selected to indicate that no assignment has been made at that position.
4. Turn SELECT to display another Soft Row position (0-9) for assignment. For each position, you can choose any effect parameter, including one(s) you have already assigned to a Soft Row position.
5. When you have arranged the Soft Row assignments as you want them, press **Edit** to exit Soft Row Assign and return to your last position in the Edit matrix. Pressing **Up** or **Down** once will also cause you to exit Soft Row Assign. Another push of either of these buttons will move you vertically through the Edit matrix.

When you return to **Go** mode, you will see the parameters in the order you assigned them — the spaces from any unassigned row positions will not appear.

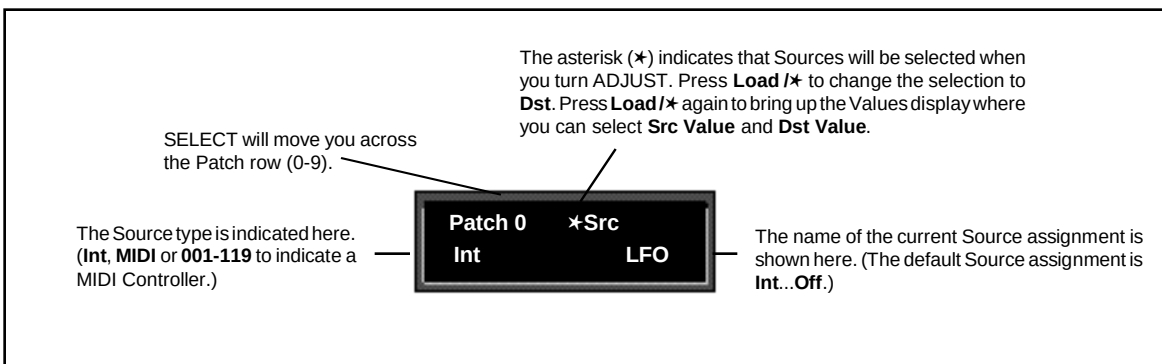
Patching Patching is the ability to assign a control (Source) to any PCM 81 parameter (Destination). This allows you to alter the value of the parameter by manipulating the control Source. For example, you can select the front panel ADJUST knob as a Source and an effect's Mix parameter as a Destination. This simple patch will allow you to dynamically alter the mix of the effect whenever you turn ADJUST. You can create as many as 10 patches, each with as many as 8 pairs of pivot points. You can patch multiple parameters to a single controller, or patch multiple Sources to a single Destination.

About Sources All Sources are the same in the sense that each generates an output value in the range of 0-127. The output value is used to increase or decrease the setting of a Destination parameter. Sources differ in the manner in which they generate an output. Some generate values continuously (they're always "on"); some generate output based on the position of a particular external MIDI controller, or an external footpedal, or footswitch. Some Sources generate output based on aspects of physical performance such as how loud, how fast, or how hard you play. The PCM 81 allows you to choose from a selection of Sources as shown for each Destination. A list of the available Sources is shown on the following page.

The Patch Row Each PCM 81 effect has an identical Patch row at the end of its Edit matrix where you can make as many as 10 patches.



To make a patch, use SELECT and the **Down** button to move down through the Edit matrix to the Patch row. A display such as the following will appear.



Pressing the **Load /★** button will change the display to allow you to sequentially select: **Src**, **Dst**, **Src Value** and **Dst Value**. Press **Up** or **Down** together with **Load /★** to backstep to the previous parameter.

Three types of sources are available: Internal, MIDI and MIDI Controller. These types are indicated in the Source list by the labels: **Int**, **MIDI**, or a number (**001-119**). Turn ADJUST to scroll through the entire list of available sources.

Assigning a Source

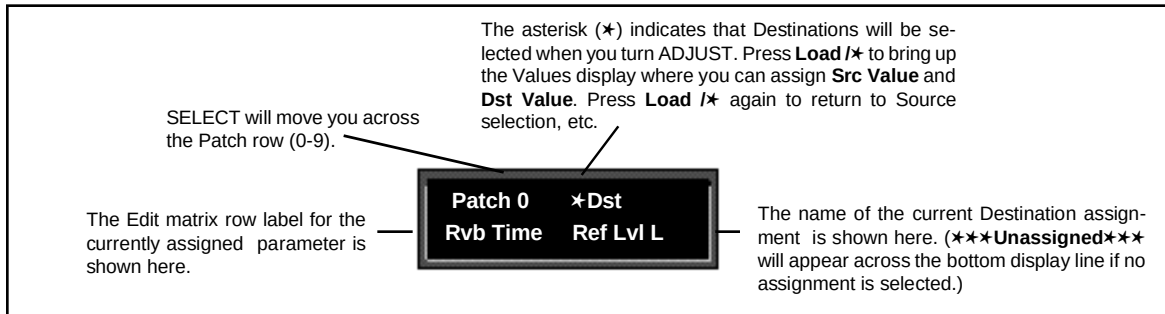
Internal
LFO
Sine
Cosine
Square
Sawtooth
Pulse
Triangle
Env L
Env R
AR Env
Latch
Sw 1
Sw 2
Sw 1 & 2
Mono Lvl
Left Lvl
Right Lvl
Footpedal
Foot Sw 1
Foot Sw 2
ADJUST
Tempo
On
Off

MIDI
P Bend
A Touch
Velocity
Last Note
Low Note
High Note
Clk Comnds

MIDI Controller Numbers			
(PCM 81 interprets 000 as Bank Select)			
001	Mod Wheel	070	Sound Var
002	Breath	071	Timbre
003	Ctl 3	072	Release
004	Foot Ctl	073	Attack
005	PortaTime	074	Bright
006	Data Entry	075	Sound 6
007	Volume	076	Sound 7
008	Balance	077	Sound 8
009	Ctl 9	078	Sound 9
010	Pan	079	Sound 10
011	Xpression	080	General 5
012	Effect 1	081	General 6
013	Effect 2	082	General 7
014	Ctl 14	083	General 8
015	Ctl 15	084	Porta Ctl
016	General 1	085	Ctl 85
017	General 2	...	...
018	General 3	090	Ctl 90
019	General 4	091	FX1 Depth
020	Ctl 20	092	FX2 Depth
...	...	093	FX3 Depth
...	...	094	FX4 Depth
031	Ctl 31	095	FX5 Depth
(PCM 81 interprets 032 as Bank Select)			
033	Ctl 33	096	Data Inc
...	...	097	Data Dec
...	...	098	NRPN LSB
063	Ctl 63	099	NRPN MSB
064	Sustain	100	RPN LSB
065	Porta On	101	RPN MSB
066	Sostenuto	102	Ctl 102
067	SoftPedal	...	...
068	Legato	119	Ctl 119
069	Hold 2		

Patch Sources

Assigning a Destination Once you have selected a Source, press **Load /★** and the display will change to allow you to assign a Destination (**Dst**).



Turn ADJUST to scroll through all of the parameters in the Edit matrix of the currently loaded effect, including the Modulation parameters.

Once you have assigned a parameter as a Destination, the controller you have assigned "owns" that parameter. Adjustments made to this parameter from the Edit matrix, will only affect the initial value of the parameter when the program is loaded.

The behavior of the parameter on program load is determined by the setting of the Patch Update parameter in Control mode (1.7). This parameter can be set to **Immediate** or **Delayed**.

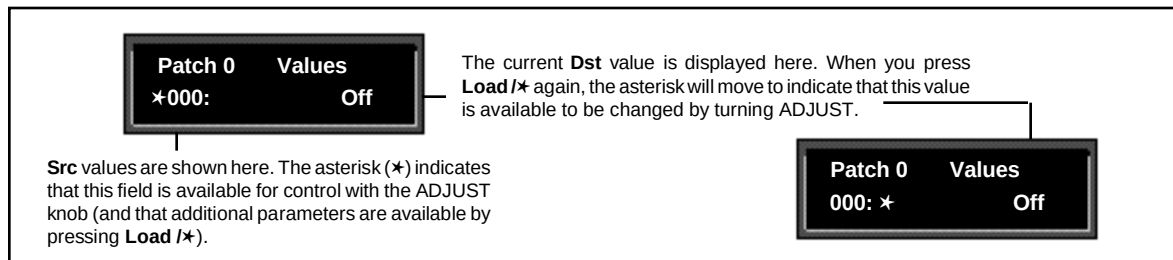
When **Immediate** is selected, the initial value of the parameter value will correspond to the controller position.

When **Delayed** is selected, the stored value of the parameter will continue to be in effect until the controller is moved. (It is, therefore, a good idea to set a sensible value to the parameter in the Edit matrix.)

Mod Row parameters that can be assigned as Patch Destinations	
LFO	Rate, Shape, P Width, Depth
AR ENV	Attack, Release, Mode
Env L	Release
Env R	Release
Sw 1	Rate, P Width, Mode
Sw 2	Rate, P Width, Mode

Assigning Values

Once you have assigned a Destination, press **Load /★** to get to the **Values** display.



This display allows you to assign Destination values to specific Source values. These assignments are made in pairs, each with a value for the Source and a value for the Destination. For example, the default is two pairs mapped as follows:

minimum Src value (0) = minimum Dst value
 maximum Src value (127) = maximum Dst value

This establishes a linear relationship between the parameter and the controller. Inverse control is accomplished easily by reversing these settings. As many as eight pairs of Destination/Source values, or pivot points, can be assigned here, providing an exciting new level of dynamic control.

Jump

When creating patches, there are situations in which you will want to leave the Patch row to adjust parameters. To make this convenient, a *Jump* command is available. Jumping is dependent on the current Patch display, and is activated simply by pressing **Edit** while a certain display is active. This will jump you out of the Patch row and to the location where you can make the necessary adjustments. Pressing **Edit** again will jump you back to the Patch Row. (Note that using any front panel controls other than those required to adjust the parameter to which you have jumped, will disable the jump. This is not catastrophic, but it will require you to return by using **Up** and **Down** and **SELECT**.) The following Jumps are available:

From the Patch row **Src** display:

- With ADJUST selected as the Source, press **Edit** to jump to the Controls row, where you can specify range limits for the ADJUST knob. Press **Edit** again to return to the Patch row.
- With any modulation parameter selected as a Source, press **Edit** to jump to the Modulation row position of the Source. For example, if the Patch source **LFO** is displayed, press **Edit** to jump to Modulation row position 0 (LFO) where you can edit any LFO parameter value. Press **Edit** again to return to the Patch row.

Jump cont'd.

From the Patch row **Dst** selection display:

- Press **Edit** to jump to the Edit controls for the parameter you have selected as the Destination. You will have complete access to all parameter controls, including any subparameters at that location. Press **Edit** again to return to the Patch row.

From the Patch row **Values** display:

- Press **Edit** to jump to the next **Src** or **Dst** value. Default values are 0...minimum, 127...maximum.

Patching Examples

The following examples illustrate how to create a patch, use the patch jump features, modify the default patch values and add an additional pivot point to the example patch values.

Creating a patch with default values

Load program **P0 0.1 EkoChorus**. Press **Edit** to enter Edit mode, then press **Up** to move to the Patch Row. Press **Load/★** until the display looks like this:



Turn SELECT to select Patch 1 (which is set to **Off**).



Turning ADJUST will scroll through the entire list of available patch Sources. Turn ADJUST counterclockwise until **LFO** is displayed in the lower right.



The LFO is now assigned as a patch Source.

Press **Load/★** to bring up patch Destinations for selection. The display should show that Destination is unassigned.



The ADJUST knob will now scroll through all of the available parameters of **EkoChorus**. The lower line of the display will show the edit matrix row label on the left, and the parameters in that row on the right.

Turn ADJUST clockwise until **FX Width** (in the effect's Controls row) is displayed in the lower right corner of the display.

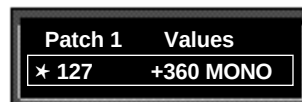


The FX Width parameter is now assigned as the patch Destination.

Now, press **Load/★** to bring up the Values display. This will show the default Destination value setting (–360 MONO). This is the value assigned to FX Width when the LFO is at its minimum value (000).



Turn ADJUST one click counterclockwise to display the default Destination value (+360 MONO) assigned to FX Width when the LFO is at its maximum value (127).



That's all there is to setting up a default patch—select a Source and Destination, and the minimum and maximum patch values are set automatically.

Of course, you will often want to modify the patch further, either by adjusting the modulation source parameters, changing the default values or adding additional pivot points. In the following sections, we'll continue using this patch to demonstrate examples of these modifications. When we're done, the new patch will add dynamic spatialization to the EkoChorus program.

Adjusting the modulation source parameters

Continuing the previous example, we'll adjust the rate of the LFO by jumping to it from the Patch row.

Press **Load/★** repeatedly to return to the Patch 1 Source selection display.



Press **Edit** to jump directly to the LFO parameters in the Mod row.

The asterisk (★) indicates that LFO **Rate** will be altered when you turn ADJUST. Press **Load /★** to change the selection to the other LFO parameters (**Shape**, **P Width** and **Depth**).

Patch Source Indicator



The display now shows position 8.0 in the Edit matrix Mod Row. The small square in the upper left corner of the display is a *patch source indicator*. This indicator appears whenever a modulator (such as the LFO in this example) has been assigned as a patch Source in the effect being edited.

Turn ADJUST to change the LFO rate to **0.10Hz**.



Now, press **Edit** to jump back to your previous position in the Patch row.



Let's modify the patch further by adjusting the Destination values to a more useful range.

Changing the default destination values

Press **Load/★** repeatedly until the Patch 1 Values screen is displayed.



Notice that the ★ is to the left of the Source value. This indicates that the Source value is selected and its value will be changed when you turn ADJUST.

Press **Load/★** once to move the ★ to the right of the Source value. When the ★ is in this position, ADJUST will change the Destination value.



The asterisk (★) indicates that the Source value will be altered when you turn ADJUST.

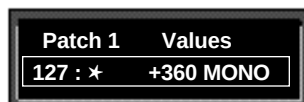


Press **Load/★** to move the asterisk to this position, where it indicates that Destination values will be altered when you turn ADJUST.

With the Destination value selected, turn ADJUST clockwise to set the value to **+0**.



Next, we'll want to adjust the Destination value when the LFO is at its maximum value. One way to do this is to press **Load/★** three times to cycle the ★ to the display of the Source value, and continue on from there — but we've provided a short cut! Press **Edit** to jump to the next assigned value (in our example, the LFO maximum value).



Note that the ★ remains in the same position, so you can just turn ADJUST to set the new Destination value. Set the value to **+90**.



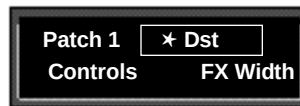
Now our example has been modified so that the LFO sweeps the FX Width Value from 0 to 90. This creates a dynamic alteration of the effect's spatial characteristics. Its stereo image changes smoothly from mono to stereo, to surround, and back again.

Adding an additional pivot point to the patch

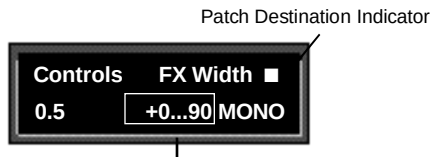
So far, our example uses only two pairs of patch values. The Destination parameter moves linearly between the value assigned at 000 and the value assigned at 127.

You can watch this change by displaying the Destination parameter. Here's how to jump directly to it from the patch:

Press **Load/★** repeatedly to return to the Patch 1 Destination selection display.



Press **Edit** to jump directly to the FX Width parameter in the Controls Row.

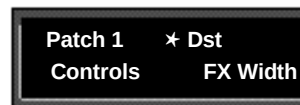


The FX Width value will be changing from 0 to 90 and back again.

The display will change to show position 0.5 in the Controls Row. Note that the value is changing continuously from 0 to 90 and back again. Notice also the small square in the upper right corner of the display. This patch destination indicator appears whenever a parameter has been assigned as a patch Destination in the effect being edited.

Now let's return to the Patch row to add a pivot point to the effect.

Press **Edit** to jump back to the Patch row.



Press **Load/★** to bring up the Values display. The last value edited will be displayed, so you will see *either* the minimum or maximum value.

Patch 1	Values
★ 000	+0 MONO

OR

Patch 1	Values
127 : ★	+90 L-R, R-L

If the ★ is not at the left of the Source value, press **Load/★** three times to move it there. (You can take a short cut instead — simultaneously press **Down** and **Load/★** to back-step.)

Patch 1	Values
★ 000	+0 MONO

OR

Patch 1	Values
★127 :	+90 L-R, R-L

Turn ADJUST to display **64**. This will be the Source value of our new pivot point. The string of dots in the destination value portion of the display indicate that there is no Destination value assigned when the source value is 64.

Patch 1	Values
★ 064 :	

Press **Load/★** to move the ★ to the right of the Source value, and turn ADJUST clockwise to set the Destination value for this point to **-45**.

Patch 1	Values
064 : ★	-45 R, L

By adding this pivot point, we have put a “kink” in the patch. The value of the Destination parameter no longer moves in a straight line between 0 and 90. Instead it moves from 0 to -45, and then from -45 to 90. This will produce a very different sounding spatial change from the original patch. You can see the difference by pressing **Load/★** twice to display the patch Destination, then pressing **Edit** to jump back to the Destination parameter to watch its value change.

Multiple Patches with the Same Destination

If you create two or more patches with the same Destination, the Destination value will be the sum of all of the patches assigned to it.

For example, if Pedal and ADJUST are both assigned to Mix, the Mix value will be the sum of the patch Destination values for those two patches. When creating multiple patches to the same Destination, you should set the individual Destinations to values which, when added together, are less than or equal to the maximum value for the parameter. Pedal and ADJUST, for example, could each have a maximum value of 50%, or they could be assigned values of 25% and 75%, 60% and 40%, etc.

When the sum of multiple patched parameter Destination values is greater than the maximum value of the parameter, the parameter value will remain at maximum until the sum of the patches falls below it.

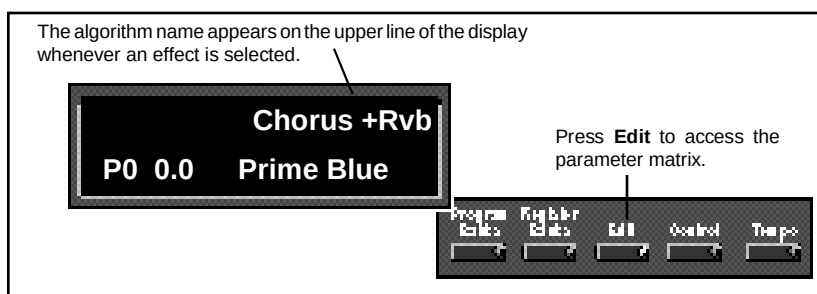
Mod Row Patches

AR ENV, **Latch**, **Sw 1** and **Sw 2** are each activated by assigning a threshold source to **T Src** that is used to turn them on and off. This assignment is a subparameter in the Mod row — not in the Patch row.

For an example, check out ADJUST and Latch in the preset **P0 3.1 Rotor Cabinet**.

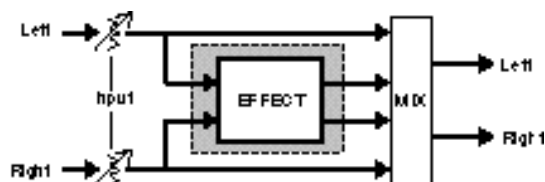
The Algorithms and Their Parameters

The PCM 81 uses 17 algorithms to create different types of effects. Each algorithm includes an uncompromised stereo reverb effect, as well as several voices of additional stereo effects. When you select any effect, the name of the algorithm from which it was derived will appear on the upper display line. Pressing **Edit** will display the last edited parameter in that algorithm's parameter matrix.



Most PCM 81 algorithms share the same general structure, shown below. The shaded area of the diagram is detailed in the individual effect descriptions that follow.

About the Algorithms



There are three general classes of algorithm: 4-Voice, 6-Voice and Pitch. The 4-Voice algorithms: Concert Hall, Plate, Chamber, Inverse and Infinite each combine a specific type of reverb with a 4-voice, general purpose stereo "effect toolbox" which we call the Reverb Shell. These algorithms also provide "post-processing" for the reverb. The 6-Voice algorithms: Glide>Hall, Chorus+Rvb, M-Band+Rvb, Res 1>Plate, and Res 2>Plate each combine a specific type of reverb with a specialized 6-voice stereo effect. The Pitch algorithms: Quad>Hall, Dual-Chmb, Dual-Plt, Dual-Inv, Stereo-Chmb, VSO-Chmb and Pitch Correct each offer a unique type of pitch shifting, combined with an uncompromised Lexicon reverb.

In this chapter, diagrams and descriptions of the Reverb Shell, and of the 4 and 6-Voice algorithms are presented first, along with pictures of each edit matrix. The diagrams are followed by a glossary of parameter descriptions, organized alphabetically by matrix row name. Within each matrix row, parameters are organized as far as possible in the sequence in which they appear in the row. The Pitch algorithms, most of which make use of a dedicated Submixer incorporated into each algorithm, are described in the following section along with operational tips on pitch shifting in general and the submixer in particular.

The 4-Voice Algorithms

The Reverb Shell

Each of the 4-Voice algorithms share a common set of controls and parameters built around one of five stereo reverb effects: Concert Hall, Plate, Chamber, Inverse and Infinite. The diagram below shows these common controls and parameters as they are structured around a reverb effect. We call this structure the Reverb Shell. (Individual reverb effects are described on the following pages.)

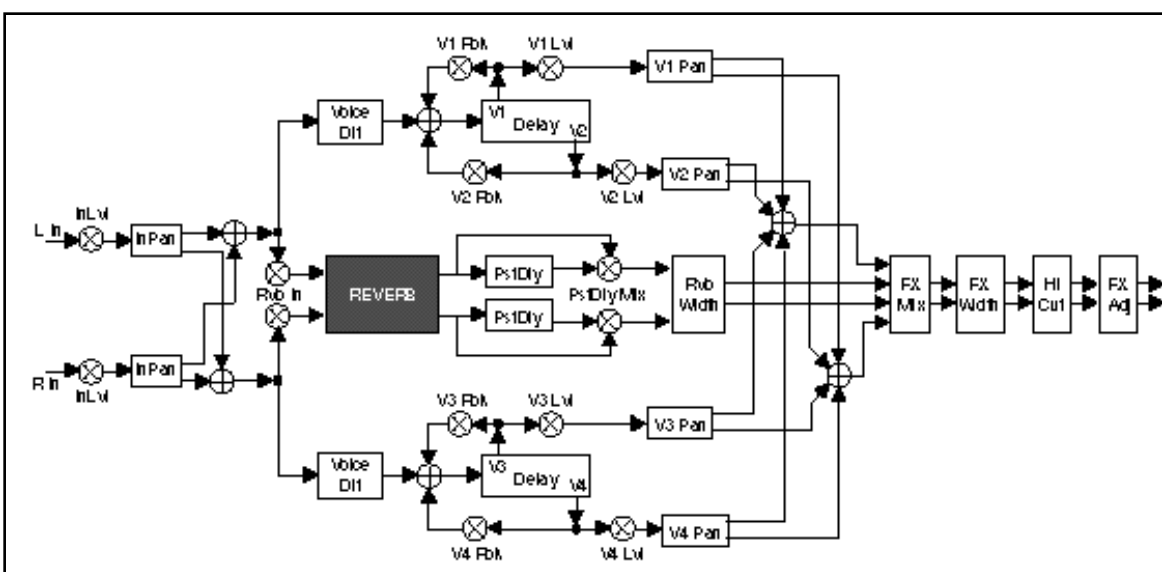
In the 4-Voice algorithms, input levels and pans determine the signal flow to left and right pairs of delay voices, and also to the reverb effect. Each delay voice has individually adjustable level, delay, feedback, and pan controls, as well as master controls for all of the voices. These algorithms can be used to create a wide variety of stereo delay and modulation effects in parallel with studio quality stereo reverberation.

The two delays after the reverb effect are referred to as post delays. The **PstDly Mix** control determines how much post delay is actually heard. The **Rvb Width** control affects only the output of the reverb portion of the effect.

The delays and the output of the reverb effect are summed at the **FX Mix** control. This control allows you to blend the relative amounts of delay and reverb.

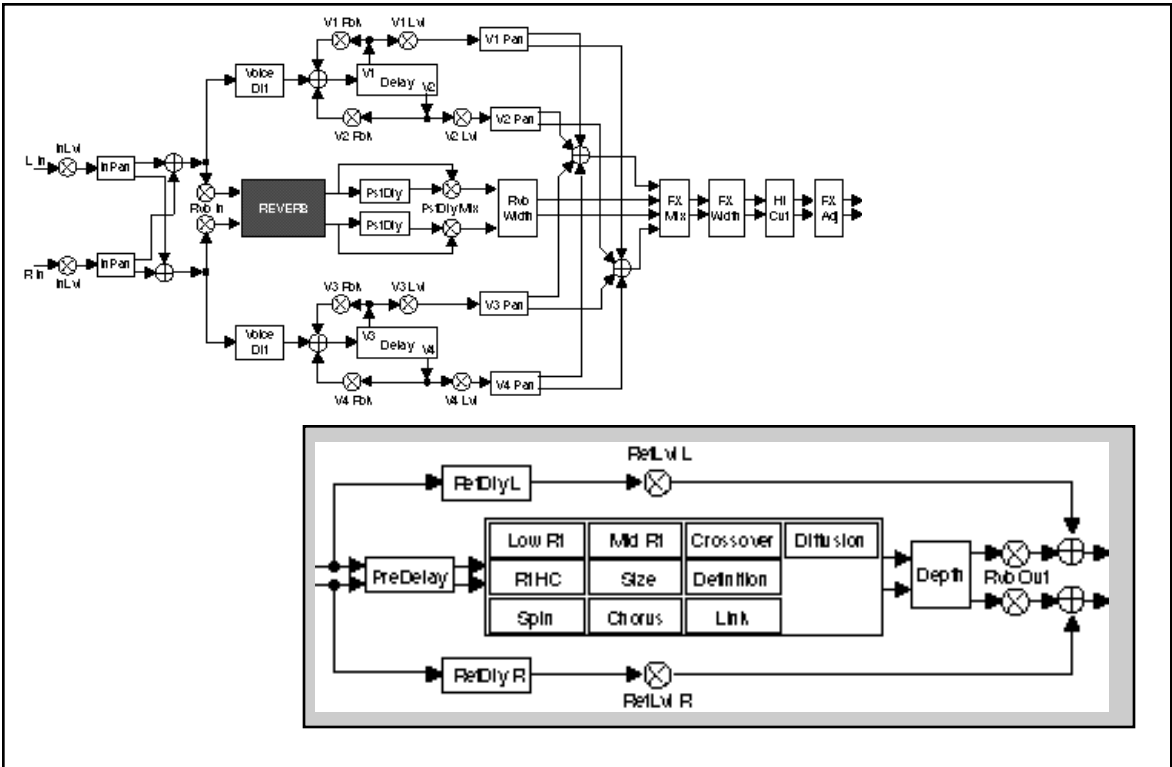
FX Width controls the spatial characteristics of the composite reverb and delay effect.

The **Hi-Cut** parameter provides high end rolloff, while **FX Adjust** determines the output volume of the entire processed signal.



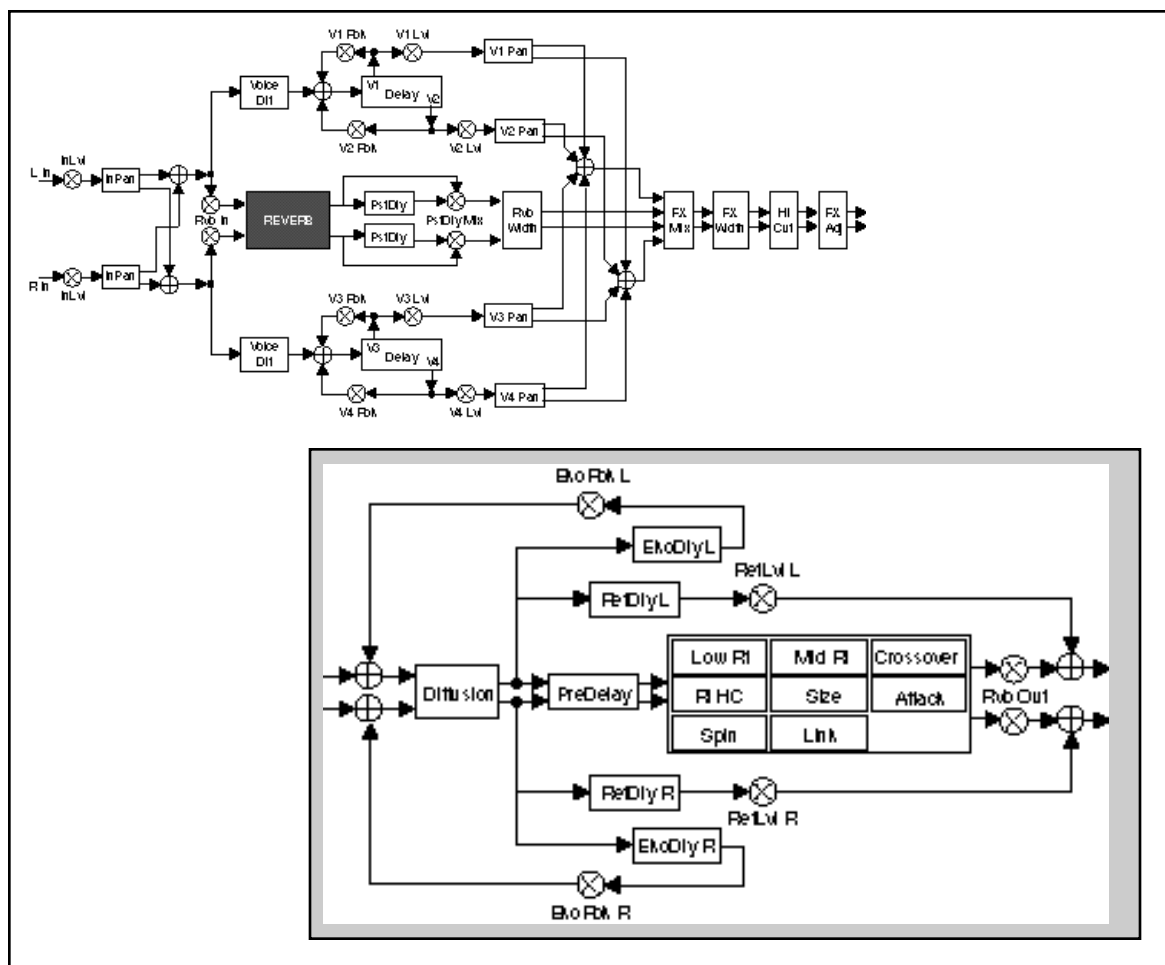
This algorithm emulates a real concert hall. The reverberation is very clean, and designed to remain behind the direct sound — adding ambience, but leaving the source unchanged. This effect has a relatively low initial echo density which builds up gradually over time.

Concert Hall



0 Control+	0.0 Mix	0.1 FXAdj	0.2 *InLvlPan	0.3 High Cut	0.4 Voice Dtl	0.5 FXMix	0.6 FX Width	0.7 *ReDlyL6T		
1 PvbTime	1.0 LowRf	1.1 Mid Rf	1.2 Crossover	1.3 Rf HC	1.4 Pre Delay	1.5 *ReDlyL6L	1.6 *Ph Gld			
2 PvbDesign	2.0 Size	2.1 Diffusion	2.2 Del	2.3 Depth	2.4 Spin	2.5 Chorus	2.6 Link	2.7 Rub Width	2.8 *Rub h Dur	
3 Level+	3.0 Master	3.1 Voice 1	3.2 Voice 2	3.3 Voice 3	3.4 Voice 4					
4 Delay Time	4.0 *Master	4.1 Voice 1	4.2 Voice 2	4.3 Voice 3	4.4 Voice 4					
5 Feedback	5.0 Master	5.1 Voice 1	5.2 Voice 2	5.3 Voice 3	5.4 Voice 4					
6 Panning	6.0 Master	6.1 Voice 1	6.2 Voice 2	6.3 Voice 3	6.4 Voice 4					
7 Modulation	7.0 Mod L RD	7.1 Mod RFR Env	7.2 Mod EnvL	7.3 Mod EnvR	7.4 Mod Lch	7.5 Mod Sml	7.6 Mod SmlC			
8 Patch++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9

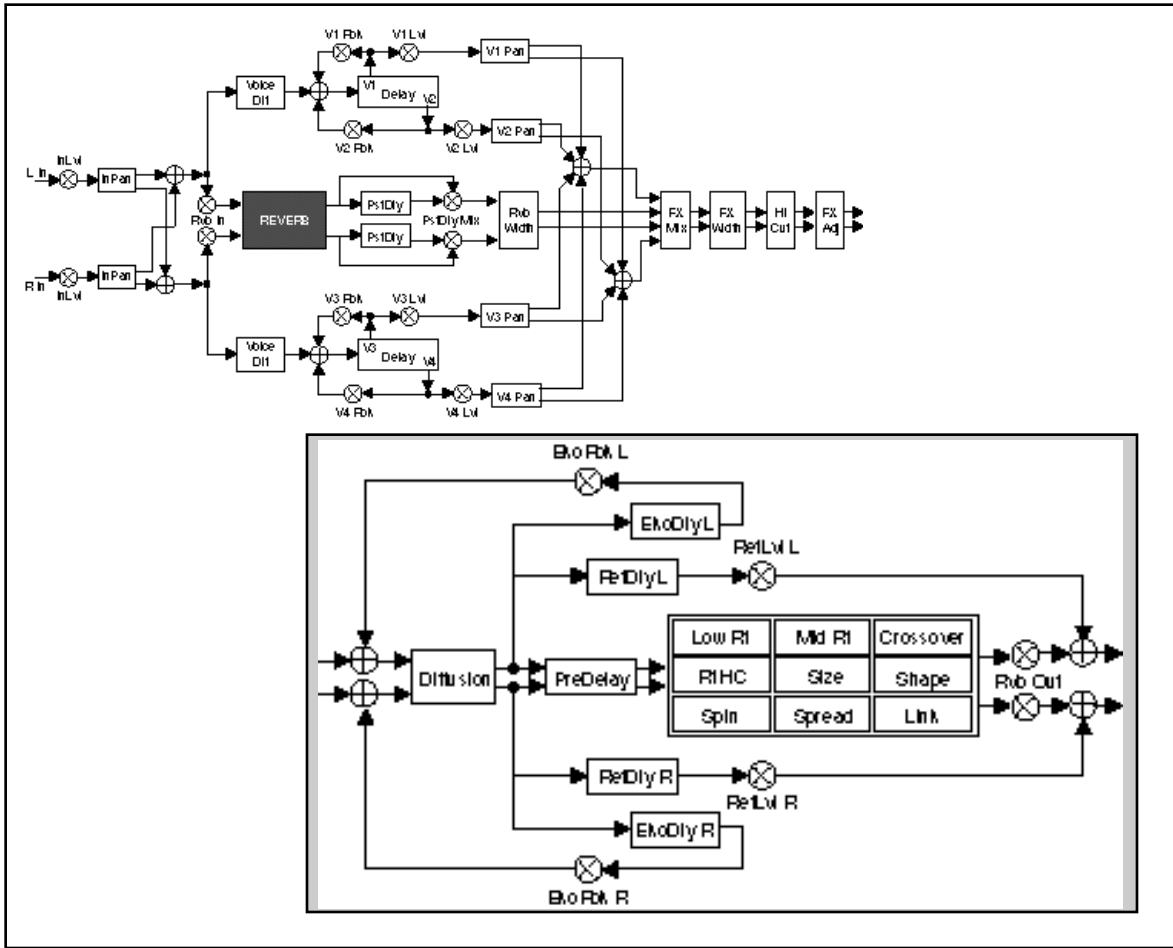
Plate The Plate algorithm mimics the sound of metal plates, with high initial diffusion and a relatively bright sound. This makes them a good choice for enhancing any type of percussion.



0 Control+	0.0 Mix	0.1 FX Adj	0.2 Initial Pan	0.3 High Cut	0.4 Voice Dtl	0.5 Fb Mix	0.6 Fb Width	0.7 Pre Dly T
1 Reverb Time	1.0 Low Rf	1.1 Mid Rf	1.2 Crossover	1.3 Rf H C	1.4 Pre Delay	1.5 Reverb Lvl L	1.6 Reverb Lvl R	1.7 Pre Gain
2 Reverb Design	2.0 Size	2.1 Diffusion	2.2 Attack	2.3 Sph	2.4 Link	2.5 Reverb Width	2.6 Reverb In Out	
3 Level+	3.0 Master	3.1 Voice 1	3.2 Voice 2	3.3 Voice 3	3.4 Voice 4			
4 Delay Time	4.0 Master	4.1 Voice 1	4.2 Voice 2	4.3 Voice 3	4.4 Voice 4			
5 Feedback	5.0 Master	5.1 Voice 1	5.2 Voice 2	5.3 Voice 3	5.4 Voice 4			
6 Panning	6.0 Master	6.1 Voice 1	6.2 Voice 2	6.3 Voice 3	6.4 Voice 4			
7 Modulation	7.0 Mod L RD	7.1 Mod RFR Env	7.2 Mod Env L	7.3 Mod Env R	7.4 Mod Lach	7.5 Mod Sml 1	7.6 Mod Sml 2	
8 Patch++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7 Patch 8 Patch 9

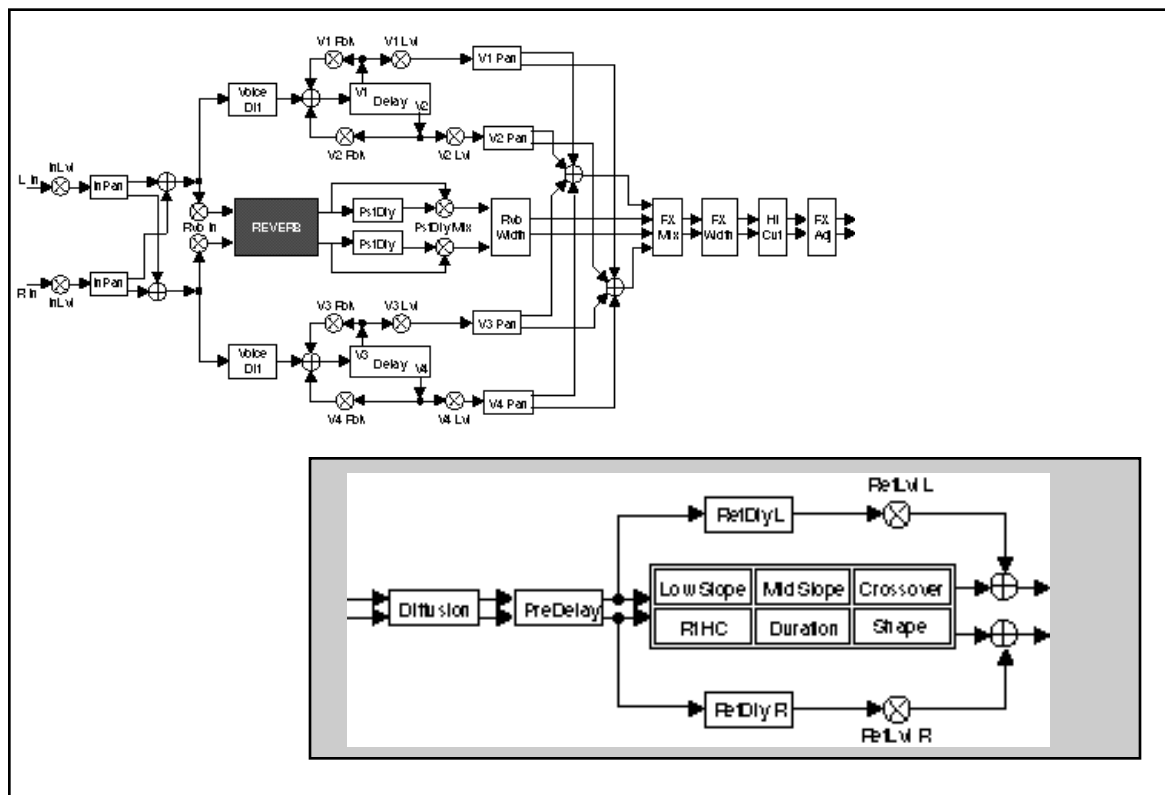
The Chamber algorithm produces an even, relatively dimensionless reverberation, with little change in color as the sound decays. The initial diffusion is similar to the Concert Hall algorithm, but the sense of space and size is much less obvious. This characteristic, along with the low color in the decay tail makes Chamber useful on a wide range of material. It is especially useful on spoken voice, giving a noticeable increase in loudness with very low color.

Chamber



0	Controls	0.0 Mix	0.1 FX Adj	0.2 RelLvlPan	0.3 High Cut	0.4 Voice DI	0.5 FX Mix	0.6 FX Width	0.7 RelLvlT
1	Rev Time	1.0 Low Rl	1.1 Mid Rl	1.2 Crossover	1.3 Rl HC	1.4 Pre Del	1.5 RelLvlL	1.6 RelLvlR	1.7 RelLvlD
2	Rev Design	2.0 Size	2.1 Diffusion	2.2 Shape	2.3 Spread	2.4 Spln	2.5 Link	2.6 Rub Width	2.7 Rub InOut
3	Level	3.0 Master	3.1 Voice 1	3.2 Voice 2	3.3 Voice 3	3.4 Voice 4			
4	Delay Time	4.0 Master	4.1 Voice 1	4.2 Voice 2	4.3 Voice 3	4.4 Voice 4			
5	Feedback	5.0 Master	5.1 Voice 1	5.2 Voice 2	5.3 Voice 3	5.4 Voice 4			
6	Panning	6.0 Master	6.1 Voice 1	6.2 Voice 2	6.3 Voice 3	6.4 Voice 4			
7	Modulation	7.0 Mod LFO	7.1 Mod RR Env	7.2 Mod EnvL	7.3 Mod EnvR	7.4 Mod Latch	7.5 Mod Sml	7.6 Mod SmlC	
8	Patch	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7
		Patch 8	Patch 9						

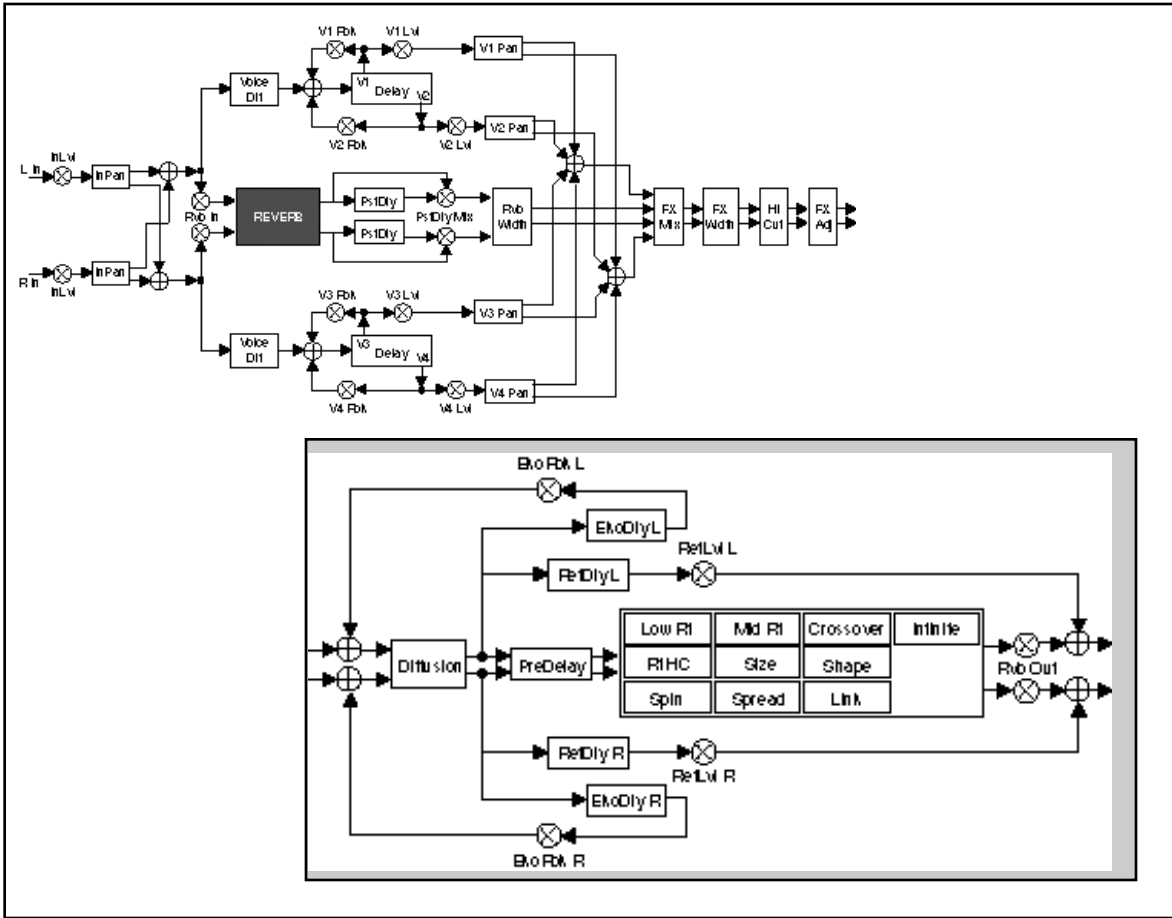
Inverse The Inverse algorithm allows you to vary the slope of the initial portion of the reverb envelope. The slope can decay, remain level, or rise over a variable time interval. When the time interval is up, the reverberation cuts off abruptly. The resulting effect is similar to a gate, but is not at all dependent on the level or complexity of the input signal. Slopes are adjustable over a negative, even, or positive slope. Positive slopes create inverse effects, while more even slopes create gated effects. Negative slope values have rather natural reverb tails.



0	Control+	0.0	Mix	0.1	FX Adj	0.2	RetLvl Pan	0.3	High Cut	0.4	Voice Dtl	0.5	FX Mix	0.6	FX Width	0.7	FX Adj
1	Reverb Time	1.0	Low Slope	1.1	Mid Slope	1.2	Crossover	1.3	RHC	1.4	Pre Delay	1.5	RetLvl L	1.6	RetLvl R		
2	Reverb Design	2.0	Duration	2.1	Diffusion	2.2	Shape	2.3	Reverb Width	2.4	Reverb In						
3	Level+	3.0	Master	3.1	Voice 1	3.2	Voice 2	3.3	Voice 3	3.4	Voice 4						
4	Delay Time	4.0	Master	4.1	Voice 1	4.2	Voice 2	4.3	Voice 3	4.4	Voice 4						
5	Feedback	5.0	Master	5.1	Voice 1	5.2	Voice 2	5.3	Voice 3	5.4	Voice 4						
6	Panning	6.0	Master	6.1	Voice 1	6.2	Voice 2	6.3	Voice 3	6.4	Voice 4						
7	Modulation	7.0	Mod LFO	7.1	Mod RFR Env	7.2	Mod Env L	7.3	Mod Env R	7.4	Mod Latch	7.5	Mod Env L	7.6	Mod Env R		
8	Patch++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9						

Infinite is acoustically similar to the Chamber algorithm, with the addition of an Infinite parameter. When this parameter is turned on, the input to the reverberator ramps off. (Note that this still allows the Reverb Shell to be utilized.) With Infinite on, the reverb tail remains constant, creating strange and useful reverb effects. When Infinite is switched off, input to the reverberator is restored, and the current running reverb time is utilized.

Infinite



0	Control+	0.0	Mix	0.1	FX Adj	0.2	ReLvl MPan	0.3	High Cut	0.4	Voice DIT	0.5	FX Mix	0.6	FX Width	0.7	ReDly U&T			
1	Rvb Time	1.0	Low Rf	1.1	Mid Rf	1.2	Crossover	1.3	Rf HC	1.4	Pre Delay	1.5	Infinite	1.6	ReLvl MDL	1.7	ReRvl MDL	1.8	ReLvl Gld	
2	Rvb Design	2.0	Size	2.1	Diffusion	2.2	Shape	2.3	Spread	2.4	Spin	2.5	Link	2.6	Rvb Width	2.7	Rvb In Cut			
3	Level+	3.0	Master	3.1	Voice 1	3.2	Voice 2	3.3	Voice 3	3.4	Voice 4									
4	Delay Time	4.0	Master	4.1	Voice 1	4.2	Voice 2	4.3	Voice 3	4.4	Voice 4									
5	Feedback	5.0	Master	5.1	Voice 1	5.2	Voice 2	5.3	Voice 3	5.4	Voice 4									
6	Panning	6.0	Master	6.1	Voice 1	6.2	Voice 2	6.3	Voice 3	6.4	Voice 4									
7	Modulation	7.0	Mod L FO	7.1	Mod Rf Env	7.2	Mod Env L	7.3	Mod Env R	7.4	Mod Latch	7.5	Mod Env L	7.6	Mod Env R					
	Patch++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9									

The 6-Voice Algorithms

Each 6-Voice algorithm is a combination of a specific 6-voice stereo effect and a specific reverb effect. These algorithms: Glide>Hall, Chorus+Rvb, M-Band+Rvb, Res1>Plate and Res2>Plate are each optimized for a particular class of audio processing effects in combination with studio quality stereo reverberation, bringing formidable processing power and flexibility to effects creation.

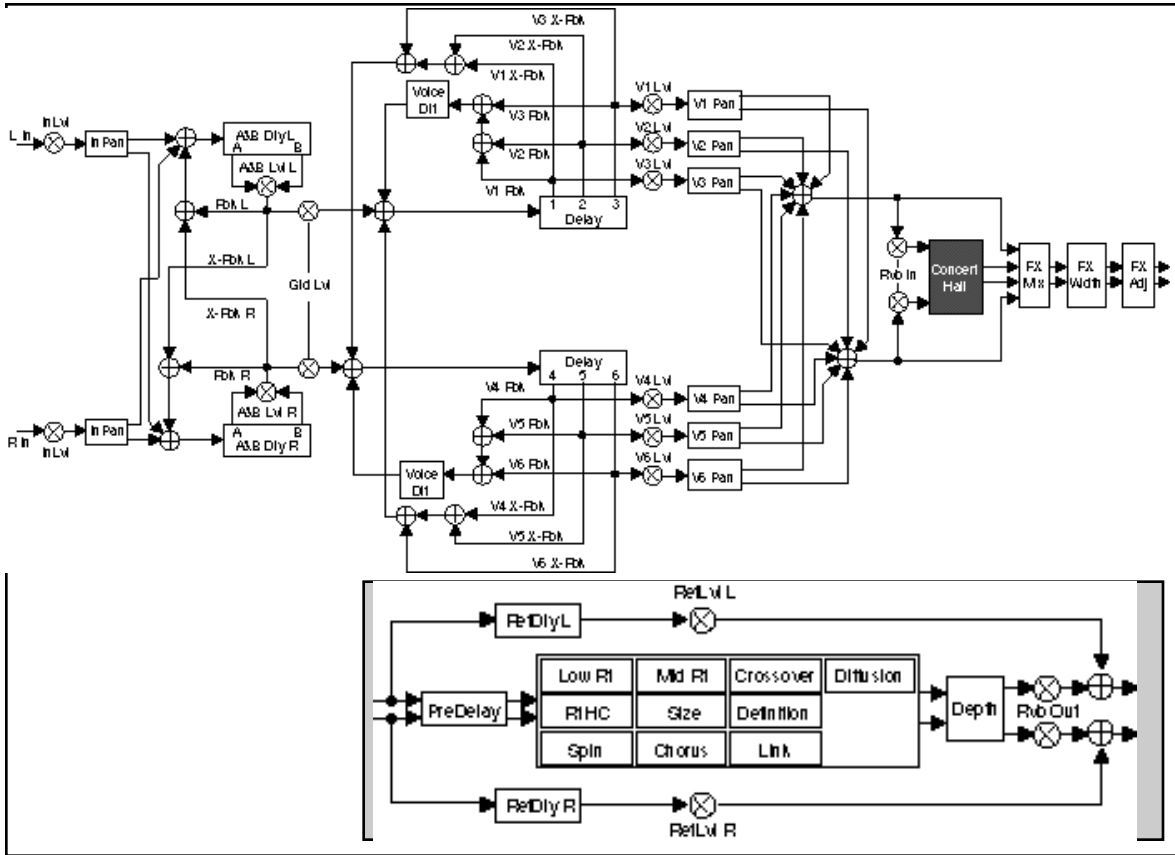
The 6-Voice algorithms use the same, uncompromised, stereo reverberation effects as the 4-Voice algorithms but, instead of the general-purpose Reverb Shell, each reverb is coupled with a unique 6-voice stereo effect: gliding delays, 6-voice chorus, multi-band EQ, and two 6-voice resonators. Voices 1-3 are connected to input audio panned to the left. Voices 4-6 are connected to input audio panned to the right. (Use the **InPan L** and **InPan R** controls at Control mode 0.2 to pan input audio.) Each voice has independent delay time, panning and level controls, in addition to other parameters specific to the particular effect.

In the Glide>Hall, Res1>Plate and Res2>Plate algorithms, the reverb effect is in series with the 6-voice effect. Use **FX Mix** to set the relative level of dry and reverberated effect.

In the M-Band+Rvb and the Chorus+Rvb algorithms, the reverb effect is in parallel with the 6-voice effect. Use **FX Mix** to set the balance of the 6-voice effect and the reverb.

A stereo pair of 2-tap gliding delays feeds six individually adjustable delay voices. Each voice has its own level, feedback, delay, cross-feedback, and pan parameters. The output of these delay voices is fed into a Concert Hall reverb algorithm. Glide>Hall is useful for creating such effects as stereo flangers, loop samplers, pitch modulation, etc. which can then be fed into the reverb.

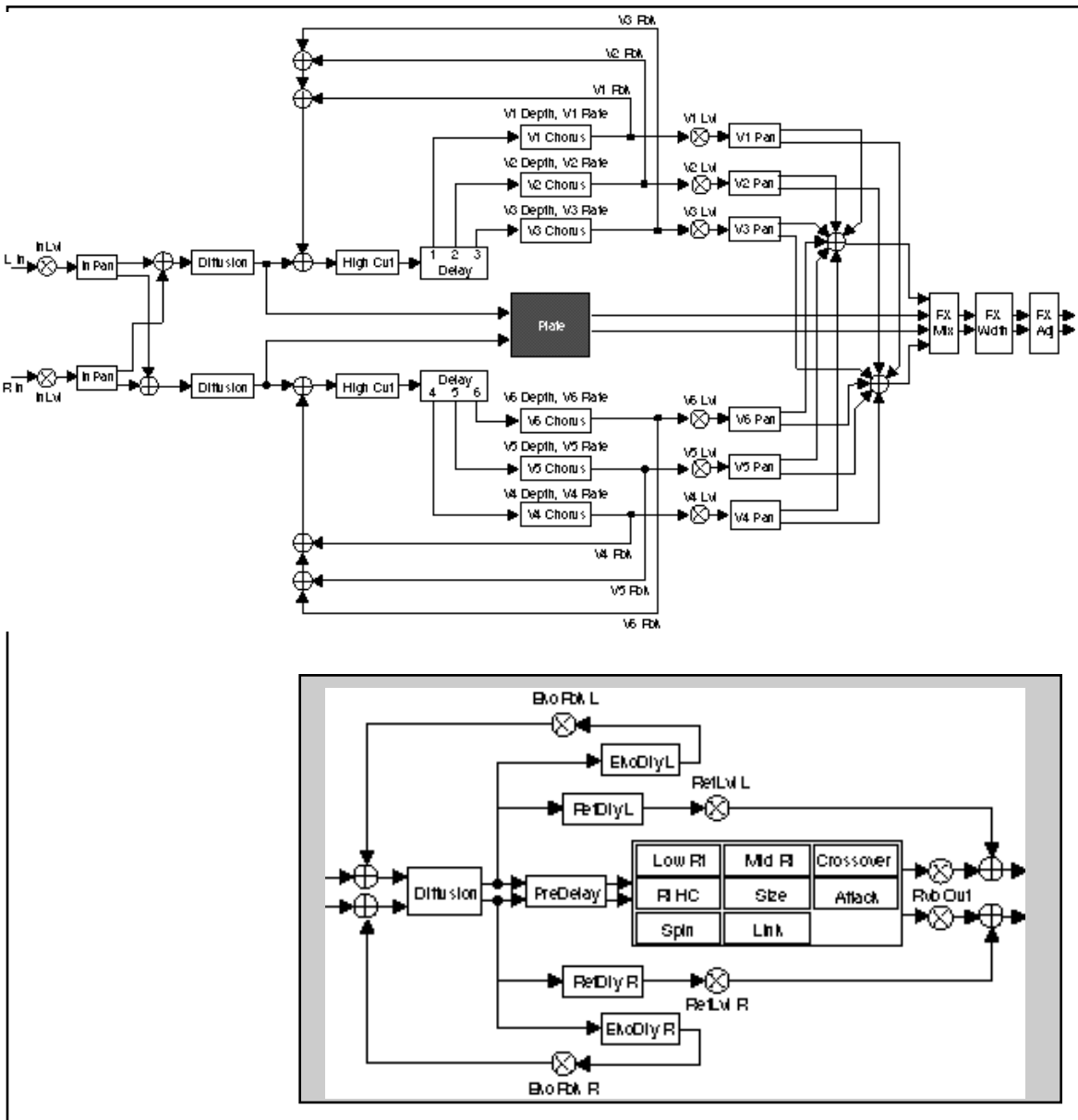
Glide>Hall



0 Control++	0.0 Mix	0.1 FX Adj	0.2 In Lvl Pan	0.3 Wake Dly	0.4 Rbk Mix	0.5 Rbk Width	0.6 Rbk Lvl
1 Rbk Time	1.0 Low R	1.1 Mid R	1.2 Crossover	1.3 RHC	1.4 Pre Delay	1.5 ReLvl Dly	
2 Rbk Design	2.0 Size	2.1 Diffusion	2.2 Del	2.3 Depth	2.4 Spin	2.5 Chorus	2.6 Link
3 Glide FX	3.0 Gld Lvl	3.1 Rbk L	3.2 Rbk R	3.3 Rbk Lvl	3.4 Rbk Rvl	3.5 Rbk Lvl R	3.6 Rbk Rvl R
4 Levels	4.0 Master	4.1 Voice 1	4.2 Voice 2	4.3 Voice 3	4.4 Voice 4	4.5 Voice 5	4.6 Voice 6
5 Delay Time	5.0 Master	5.1 Voice 1	5.2 Voice 2	5.3 Voice 3	5.4 Voice 4	5.5 Voice 5	5.6 Voice 6
6 Feedback	6.0 Master	6.1 Voice 1	6.2 Voice 2	6.3 Voice 3	6.4 Voice 4	6.5 Voice 5	6.6 Voice 6
7 Panning	7.0 Master	7.1 Voice 1	7.2 Voice 2	7.3 Voice 3	7.4 Voice 4	7.5 Voice 5	7.6 Voice 6
8 Modulation	8.0 Mod LFO	8.1 Mod Rbk Env	8.2 Mod EnvL	8.3 Mod EnvR	8.4 Mod Lch	8.5 Mod Rbk L	8.6 Mod Rbk R
9 Patch++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6
	Patch 7	Patch 8	Patch 9				

Chorus+Rvb The Chorus effect has six separately adjustable voices — allowing the PCM 81 to sound like a rack of six digital delay boxes. Each voice has its own independently adjustable chorus depth and rate, level control, delay time, feedback and panning control. The 6-voice chorus is in parallel with a plate algorithm, providing two independent stereo effects.

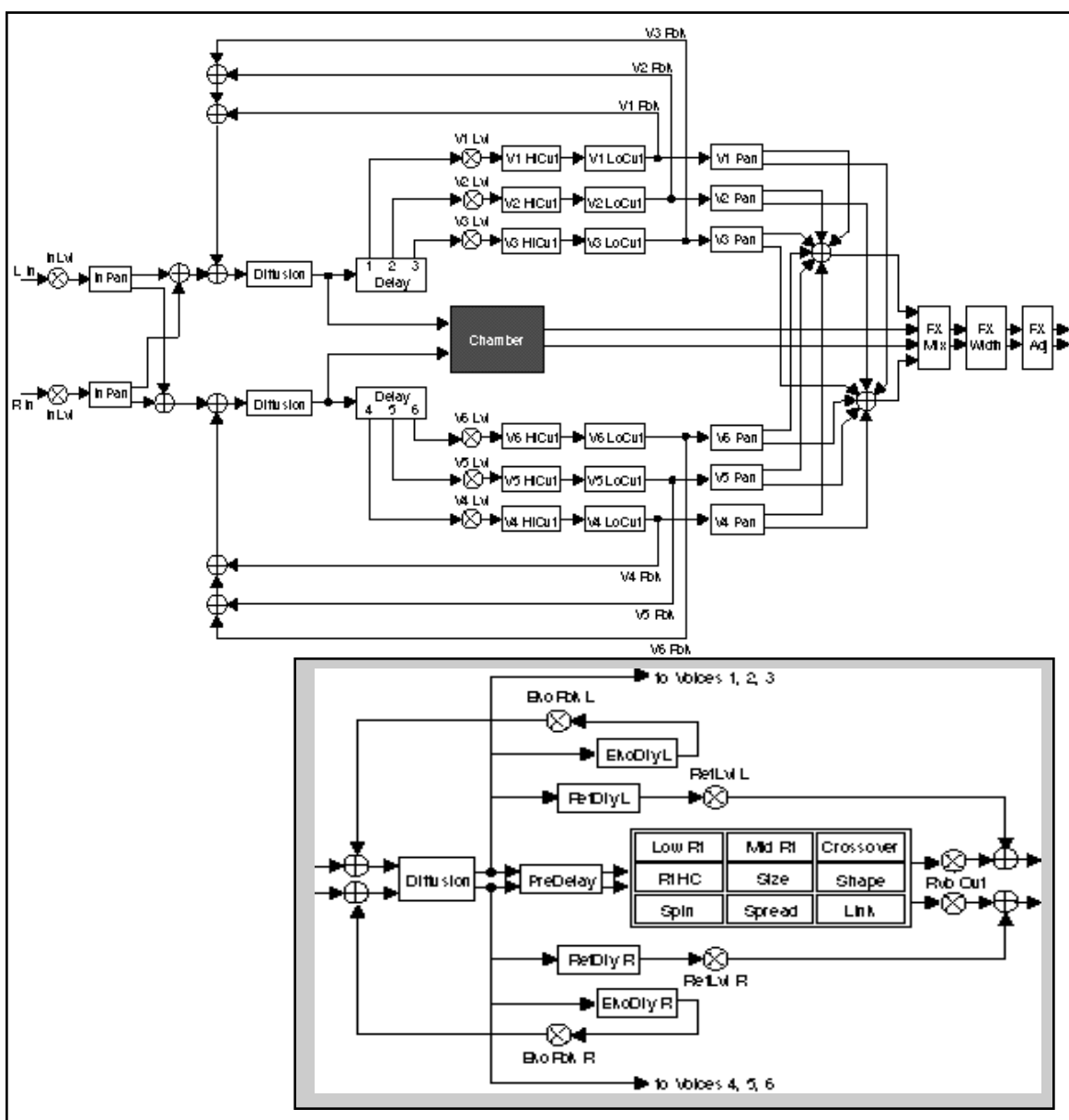
Note that the Diffusion parameter (Rvb Design 2.1) is shared by both the reverb and the chorus effect.



Chorus+Rvb cont'd.

0	Control+	0.0	Min	0.1	FX AdjUn	0.2	RelM Pan	0.3	High Cu	0.4	FX Min	0.5	FX Midh	0.6	FX DUBT
1	Pub Time	1.0	LowRt	1.1	MidRt	1.2	Crossover	1.3	Rt H C	1.4	Pre Dela.	1.5	Rel Lnt DL	1.6	BrRntDL
2	Pub Design	2.0	60%	2.1	Diffusion	2.2	Reck	2.3	Spin	2.4	Unk	2.5	Pub Out		
3	Level+	3.0	Master	3.1	Voice 1	3.2	Voice 2	3.3	Voice 3	3.4	Voice 4	3.5	Voice 5	3.6	Voice 6
4	Delay Time	4.0	Master	4.1	Voice 1	4.2	Voice 2	4.3	Voice 3	4.4	Voice 4	4.5	Voice 5	4.6	Voice 6
5	Chorus+	5.0	Master	5.1	Voice 1	5.2	Voice 2	5.3	Voice 3	5.4	Voice 4	5.5	Voice 5	5.6	Voice 6
6	Feedback	6.0	Master	6.1	Voice 1	6.2	Voice 2	6.3	Voice 3	6.4	Voice 4	6.5	Voice 5	6.6	Voice 6
7	Panning	7.0	Master	7.1	Voice 1	7.2	Voice 2	7.3	Voice 3	7.4	Voice 4	7.5	Voice 5	7.6	Voice 6
8	Modulation	8.0	Mod L FD	8.1	Mod RR Env	8.2	Mod EnvL	8.3	Mod EnvR	8.4	Mod L Arch	8.5	Mod EnvL	8.6	Mod EnvC
	Patch++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9				

M-Band+Rvb This effect features six separately adjustable voices, each with its own level control, delay time, low and high frequency filters, feedback and pan controls. The multi-band effect is in parallel with a Chamber effect, providing independent stereo effects. Note that the Diffusion parameter (Rvb Design 2.1) is shared by both the multi-band and reverb effects. Note also that, in this particular algorithm, the diffuser is within the feedback paths of the multi-band voices. This allows you to create filtered echoes that grow more diffuse with each repeat, or to create effects with filtered echoes passing through the reverberator.



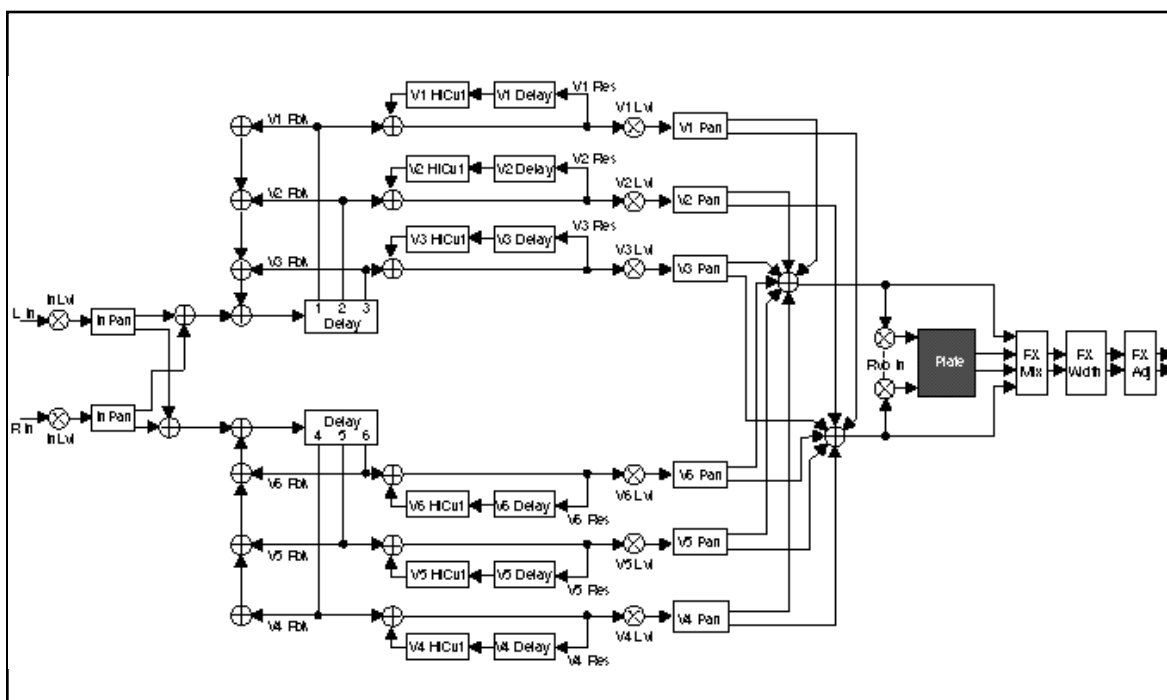
M-Band+Rvb cont'd.

0 Control+	0.0 Min	0.1 FXAdjW	0.2 *RelLWIPan	0.3 High Cut	0.4 Voice Cl	0.5 FXMin	0.6 FX Width	0.7 *PDUST				
1 Pwb Time	1.0 LowR	1.1 Mid R	1.2 Crossover	1.3 R HC	1.4 Pre Del.	1.5 *RelLWIDL	1.6 *BioRWDL	1.7 *PwIGid				
2 Pwb Design	2.0 Size	2.1 Diffusion	2.2 Shape	2.3 Spread	2.4 Spin	2.5 Link	2.6 Pwb Width	2.7 *Pwb InOut				
3 Level+	3.0 Master	3.1 Voice 1	3.2 Voice 2	3.3 Voice 3	3.4 Voice 4							
4 Delay Time	4.0 Master	4.1 Voice 1	4.2 Voice 2	4.3 Voice 3	4.4 Voice 4							
5 Feedback	5.0 Master	5.1 Voice 1	5.2 Voice 2	5.3 Voice 3	5.4 Voice 4							
6 Panning	6.0 Master	6.1 Voice 1	6.2 Voice 2	6.3 Voice 3	6.4 Voice 4							
7 Modulation	7.0 Mod L RD	7.1 Mod RFR Env	7.2 Mod EnvL	7.3 Mod EnvR	7.4 Mod Lch	7.5 Mod Sm L	7.6 Mod Sm C					
Patch++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9		

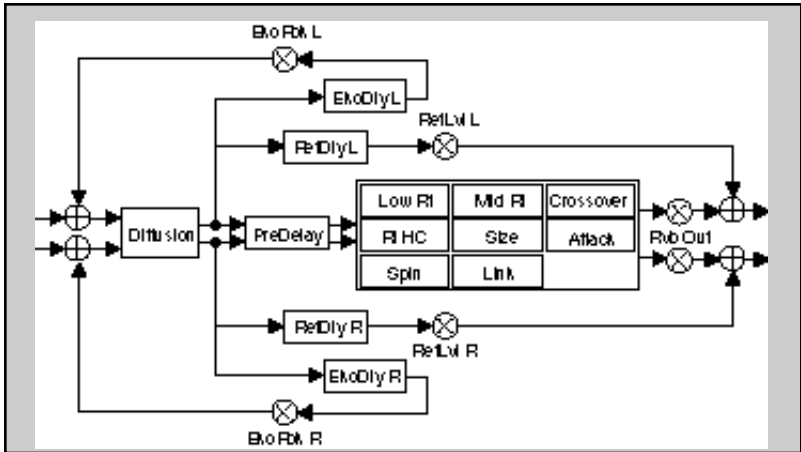
The Resonant Chord**Algorithms:
Res1>Plate and
Res2>Plate**

The Resonant Chord effects use impulsive energy at the inputs to excite six resonant voices (notes). The level, pitch, duration, and high-frequency cutoff of the overtones for each voice are separately controllable. Each voice can be panned independently. The voices resonate to some degree with any input, but the most effective excitation contains all frequencies, like percussion. Other instruments may give a quality of tonal ambience in which certain notes rise ethereally from the background. The output of the resonator is then fed into a stereo plate reverb effect.

The two algorithms differ in the way pitches are assigned to the resonators. In Res1>Plate, pitches are assigned to the six voices chromatically, in a round-robin. If, for example, MIDI note numbers are used to assign pitch, the resonators will constantly be re-tuned to the pitches of the last six MIDI notes received. (This can produce an effect similar to playing a piano with the sustain pedal depressed.) In Res2>Plate, pitches are assigned to the six resonators diatonically — harmonized with the key, scale, and root of your choice. If MIDI note numbers are used to assign pitch, the resonators will constantly be re-tuned to harmonize with the incoming notes.



The Resonant Chord Algorithms:
Res1>Plate and
Res2>Plate cont'd.



Res1>Plate

0 Controls+	0.0 Mix	0.1 FX Adjw	0.2 RelLvl Pan	0.3 FX Mix	0.4 FX Width	0.5 RelLvl T				
1 Pub Time	1.0 LowRf	1.1 Mid Rf	1.2 Crossover	1.3 Rf HC	1.4 Pre Delay	1.5 RelLvl DL	1.6 B.o.Rk DL			
2 Pub Design	2.0 Size	2.1 Diffusion	2.2 Attack	2.3 Sptn	2.4 Link	2.5 RelLvl Du				
3 Levels	3.0 Master	3.1 Voice 1	3.2 Voice 2	3.3 Voice 3	3.4 Voice 4	3.5 Voice 5	3.6 Voice 6			
4 Delay Time	4.0 Master	4.1 Voice 1	4.2 Voice 2	4.3 Voice 3	4.4 Voice 4	4.5 Voice 5	4.6 Voice 6			
5 Performance	5.0 MasterHC	5.1 Voice 1	5.2 Voice 2	5.3 Voice 3	5.4 Voice 4	5.5 Voice 5	5.6 Voice 6			
6 Pitch	6.0 Assign	6.1 Tuning	6.2 Active	6.3 Unison						
7 Panning	7.0 Master	7.1 Voice 1	7.2 Voice 2	7.3 Voice 3	7.4 Voice 4	7.5 Voice 5	7.6 Voice 6			
8 Modulation	8.0 Mod L FO	8.1 Mod RFR Env	8.2 Mod EnvL	8.3 Mod EnvR	8.4 Mod Lach	8.5 Mod EnvL	8.6 Mod EnvR			
Patches++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9

Res2>Plate

0 Controls+	0.0 Mix	0.1 FX Adjw	0.2 RelLvl Pan	0.3 FX Mix	0.4 FX Width	0.5 RelLvl T				
1 Pub Time	1.0 LowRf	1.1 Mid Rf	1.2 Crossover	1.3 Rf HC	1.4 Pre Delay	1.5 RelLvl DL	1.6 B.o.Rk DL			
2 Pub Design	2.0 Size	2.1 Diffusion	2.2 Attack	2.3 Sptn	2.4 Link	2.5 RelLvl Du				
3 Levels	3.0 Master	3.1 Voice 1	3.2 Voice 2	3.3 Voice 3	3.4 Voice 4	3.5 Voice 5	3.6 Voice 6			
4 Delay Time	4.0 Master	4.1 Voice 1	4.2 Voice 2	4.3 Voice 3	4.4 Voice 4	4.5 Voice 5	4.6 Voice 6			
5 Performance	5.0 Master HC	5.1 Voice 1	5.2 Voice 2	5.3 Voice 3	5.4 Voice 4	5.5 Voice 5	5.6 Voice 6			
6 Pitch	6.0 Master	6.1 Voice 1	6.2 Voice 2	6.3 Voice 3	6.4 Voice 4	6.5 Voice 5	6.6 Voice 6			
7 Panning	7.0 Master	7.1 Voice 1	7.2 Voice 2	7.3 Voice 3	7.4 Voice 4	7.5 Voice 5	7.6 Voice 6			
8 Modulation	8.0 Mod L FO	8.1 Mod RFR Env	8.2 Mod EnvL	8.3 Mod EnvR	8.4 Mod Lach	8.5 Mod EnvL	8.6 Mod EnvR			
Patches++	Patch 0	Patch 1	Patch 2	Patch 3	Patch 4	Patch 5	Patch 6	Patch 7	Patch 8	Patch 9

The Parameters

PCM 81 parameters are organized into labeled rows within each edit matrix. Although there are similarities among all matrixes, such as having a row of Controls first, and Modulation and Patching rows last, some of the parameters within each row, and some entire rows are unique to specific algorithms.

This section contains descriptions for all PCM 81 parameters, organized alphabetically by row label as follows:

Chorus
Controls
Delay Time
Feedback/Cross-Feedback
Filters
Glide FX
Levels
Modulation
Panning
Patching
Pitch
Resonance
Rvb Design
Rvb Time

Individual parameter descriptions within each row are presented, as far as possible, in the order in which they appear from left to right in the edit matrix.

Chorus

In the Chorus+Rvb algorithm, Row 5 provides parameters which provide rate and depth controls for each of the six chorus voices, as well as master rate and depth controls.

MstDepth and MstRate

Two Master Chorus parameters in the first row position allow Chorus, Depth and Rate for all voices to be simultaneously scaled. Each control has a range of 0-200%. Press **Load/★** to toggle between MstDepth and MstRate.

Voices (Depth and Rate)

Each voice position has Depth and Rate controls to set the time range (delay) and rate of change across that time range (pitch) of the chorusing effect. (Press **Load/★** to toggle between them.) Depth provides settings of 0-500ms in single millisecond increments. Rate parameter can be set to 0Hz (Off), or to one of one-hundred selections from 0.01-3.50Hz. Depths of 10-30ms combined with Rates as high as 0.50Hz provide subtle chorusing and multivoicing effects. Depths of hundreds of milliseconds combined with higher Rates provide a wide range of pitch shifting effects.

Row 0 of every algorithm contains parameters that provide overall control of both the reverb and voice effects. **Controls**

Mix

Mix controls the ratio of dry and wet signal present at the PCM 81 outputs. When the PCM 81 is patched into a console or an instrument amplifier through an auxiliary or effects loop, this control should always be set to 100% wet. (Control Mode 1.1 allows you to select a global Mix setting.) When an instrument is plugged directly into the PCM 81, a setting between 45 and 65% is a good starting point.

FX Adjust

FX Adjust controls the level of the processed (wet) signal before it is mixed with the dry signal (if any) and sent to the audio outputs. The range of this parameter is +12dB to -73dB, and Off (wet output muted.) FX Adjust is inactive when the PCM 81 is in Bypass mode, so it is a good idea to compare the levels with Bypass on and off while editing this parameter. Adjust it until the audio level sounds the same with bypass on and off.

InLvl L&R, InPan L&R

InLvl L and InLvl R control the level and phase of the unprocessed dry signal into the effects. The range of these parameters is from Full (0dB) Phase Normal, to -85dB Phase Normal, Off, -85dB Phase Inverted to Full (0dB) Phase Inverted.

InPan L and InPan R control the location of the left and right input audio as a stereo image to the effect's audio input. Unmodified stereo imaging to the effects can be achieved by setting InPan L hard left (50L) and InPan R hard right (50R). Changing InPanL to any other value (49L to 50R), takes the left audio input and presents it as a panned source to the effect's left and right input. Changing InPan R to any value other than 50R, does the same for the right audio input.

High Cut

High Cut sets the high frequency cutoff of a low-pass filter. This parameter affects both channels.

Voice Dif

This parameter is available in all of the 4-Voice Algorithms, and in Glide>Hall. It controls the density of echoes in the delay voices, independently of Reverb diffusion. High levels of diffusion thicken, or smear, the echoes. This is most noticeable on material with sharp transients.

FX Mix

FX Mix controls the balance of reverb and non-reverb effects. (Non-reverb effects are those processed as "voices" controlled by level, delay and feedback parameters generally found in rows 3-7.) The range of this control is 0 to 100%.

Controls cont'd. FX Width

FX Width can be thought of as an extension of typical mono to stereo imaging controls. The range of this parameter is -360 to +360, in single digit increments. Values of -360, 0, or +360 cause the effect's audio output to be mono. Values of -315 and +45 cause the output to be normal left/right stereo. Values of -45 and +315 cause "swapped", or right/left stereo.

Value	Display Label	Description
360	MONO	Phase Normal Mono
315	R, L	Phase Normal R/L stereo
270	R-L, L-R	R-L, L-R surround
225	STEREO INV	Phase Inverted L/R Stereo
180	MONO INV	Phase Inverted Mono
135	R, L INV	Phase Inverted R/L Stereo
90	L-R, R-L	L-R, R-L surround
45	STEREO	Phase Normal L/R stereo
0	MONO	Phase Normal Mono
-45	R,L	Phase Normal R/L Stereo
-90	R-L, L-R	Phase Inverted R-L, L-R surround
-135	STEREO INV	Phase Inverted L/R Stereo
-180	MONO INV	Phase Inverted Mono
-225	R, L INV	Phase Inverted R/L Stereo
-270	L-R, R-L	L-R, R-L surround
-315	STEREO	Phase Normal L/R Stereo
-360	MONO	Phase Normal Mono

The large and duplicated range of this parameter allows smooth glides from any Left/Right Mix, Phase, Mono/Stereo image point to any other. These are on the output of the FX Mix. Of particular interest are: 0 MONO, 45 STEREO, and 90 L-R, R-L (surround channel). This parameter can be changed in real-time for fascinating spatial effects.

ADJUST

This control allows you to specify the range of control of the ADJUST knob when it is patched to a parameter. Low and High settings are available, allowing you to select full range (0-127) for those parameters where it is appropriate and to limit the range in other cases. You can, for example set the Low value to 0, and the High value to 1, to have ADJUST behave as an on/off control. Many of the presets take advantage of this, and have ADJUST range limits which are appropriate to the specific parameter.

In each algorithm, the Delay Time row contains parameters for delay settings of each voice, as well as master delay parameters for all voices.

Delay Time

Master, GldResp, GldRange, Clear

Press **Load/★** to cycle through selections: **Master**, **GldResp**, **GldRange** and **Clear**.

Master

This control allows you to simultaneously change the delay times of all voices in the effect. The available range is from 0-200%. This provides a simple way to expand or close in all of the voice delay times. If a voice's delay time is set to 500ms, changing the setting of the Master parameter to 200% will change the delay time to 1000ms. Setting this parameter to 0% is an easy way to set all the delays to 0 from a single control.

GldResp

In all of the 4-Voice algorithms, and in the Chorus+Rvb algorithm, this parameter controls the response of the glide. The range is 0-100, with a typical default setting of 50. This setting provides a good, real-time glide. Changing a delay time from 0 to 100ms with a glide response of 0, causes the glide delays to take a minute or more, allowing subtle changes in delay times and echoes with no noticeable pitch shifting. A glide response of 100 is ultrafast, causing high speed pitch shift, and even chirping effects with large delay time changes.

GldRange

In all of the 4-Voice algorithms, and in the Chorus+Rvb algorithm, this parameter controls the range over which changes to the voice delay time will be performed as a glide. For example, if GldRange is set to 100ms, and the PstDly L and R times are varied over a range of 100ms, those delays will glide smoothly from one delay time to another. Suddenly changing the delay time to a greater value, such as 200ms, will cause the audio delay to change instantly to the new time without gliding. The available range is 0-1.365sec, with 0 causing all delay time changes to be "instantaneous", and 1.365 causing all delay time changes to glide.

Clear

This is an On/Off control which allows you to instantly clear all of the audio in the delay voices. A typical use for this control might be to patch it to a footswitch to allow one tap removal of all old audio from the delays to start fresh. While Clear is on, no new audio will pass through any delay voice set above 1ms.